

**ADMINISTRATIVE INFORMATION****FMI No: 25334****FMI Title: Performix Tube Field Inspection**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety ☐ Mandatory ☐ Mandatory on Request	GEMS-AM: GEMS-E: GEMS-A:	Issue Date: Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole. ☒ Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole. ☐ Parts Ordering: GEMS - Americas, contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator ☐ No Parts are required to complete this FMI.
☐ Batch with	FMI # To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

"[Preliminary Steps](#)" MUST be reviewed prior to any activity on this FMI.

Unit Tracked:**FMI Instructions Part No.: 2399874-100****Do NOT leave unsecured on site.**



Completion Certification Report

Mail To

GEMS - Americas

GE Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe

GEMS - E / ESC
FMI Admin. 322
BP34
F 78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25334 FMI Title: Performix Tube Field Inspection

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code:

- 1.) Modification Complete or previously modified.
- 2.) FMI not required per FMI instruction.
- 3.) Not modified. Unit in storage. FMI kit returned to stock.
- 4.) Modification refused or equipment altered. Customer signature and title required.
- 5.) Unit location unknown or unit not in area. Product Locator notified.

**Customer Refusal or Equipment Altered
- Modification Pending.**

Installer
Comments

Effectivity

All of the following CT Systems, with the defined gantry types, are affected by this FMI:

- LightSpeed 2X (QX/I, Plus)
- LightSpeed 3X (QX/I, Plus, Ultra)
- LightSpeed 4X (16 Slice)
- HiSpeed QX/i
- Discovery LS (PET/CT)
- Discovery ST (PET/CT)

The following LightSpeed Family Gantry Model Types are affected by this FMI:

- | | |
|-------------|-------------|
| • 2137130 | • 2281177-3 |
| • 2137130-5 | • 2281177-4 |
| • 2234921-3 | • 2305500 |
| • 2260619 | • 2339985 |
| • 2260619-3 | • 2341799 |
| • 2281177 | • 2362605 |
| • 2281177-2 | • 5101272 |

Purpose

The Purpose of this FMI is to perform X-Ray Tube Casing and X-Ray Tube Heat Exchanger inspections on LightSpeed (2X, 3X, 4X), HiSpeed QX/i and Discovery LS/ST systems that operate with a CT Gantry rotation speed capability of 0.5 second/rev. The field inspections are required per Product Safety Reports:

- PSR 13010961, Performix X-Ray Tube
- PSR 13010989, Performix Tube Heat Exchanger

Mandatory On Request FMI 25335 has been established to perform additional evaluation of Tubes that are identified per the Tube Casing inspection of this FMI.

There is a Data sheet with this FMI that must be filled out and returned with the results of the inspection.



NOTICE

If the site is seeing Image Quality artifacts since the last tube change, please refer to [“Appendix C - Tube Bolt Information,”](#) beginning on [page 29](#). If you have further questions, call the Online Support Center for direction.



NOTICE

IMPORTANT: You **MUST** review all the training videos and information available on the supplied CD-ROM prior to performing this FMI.

- Torque Wrench Use for Tube Mounting
- Heat Exchanger Test FMI 25334
- Performix Tube FMI 25334
- Color Pictures reference document

Furnished Materials

The following parts are provided with this FMI

Qty	Description	Part #
1	FMI document	2399874-100
1	Tube and Service Information CD-ROM	2399977
1	LightSpeed Tube Bolt kit	2391264-2
1	Tube Label, Passed FMI 25334/25335	2400473
2	Tube Label, Red - Defective Sticker	2358146
1	Tube Label, Tube requires FMI 25335	2401457

Required Tools

The following hardware tools are required to perform this FMI:

Qty	Description	Part #
1	Small mirror, hand held	Field Supplied
1	Torque Wrench Kit	<i>Per GEMS Service Regional Team</i> 46-268521G1 (Americas)
1	Tube Hoist	<i>Per GEMS Service Regional Team</i> Arm: 2282509, Boom: 2282510 (Americas)

The following Service Documentation CD-ROMs are needed to complete this FMI. The FMI instruction will identify when these documents are to be used

Qty	Description	Part #
1	LightSpeed Series Service Information CD-ROM	2360027-200 Rev 03
1	Discovery Series Service Information CD-ROM	2371120

People Power

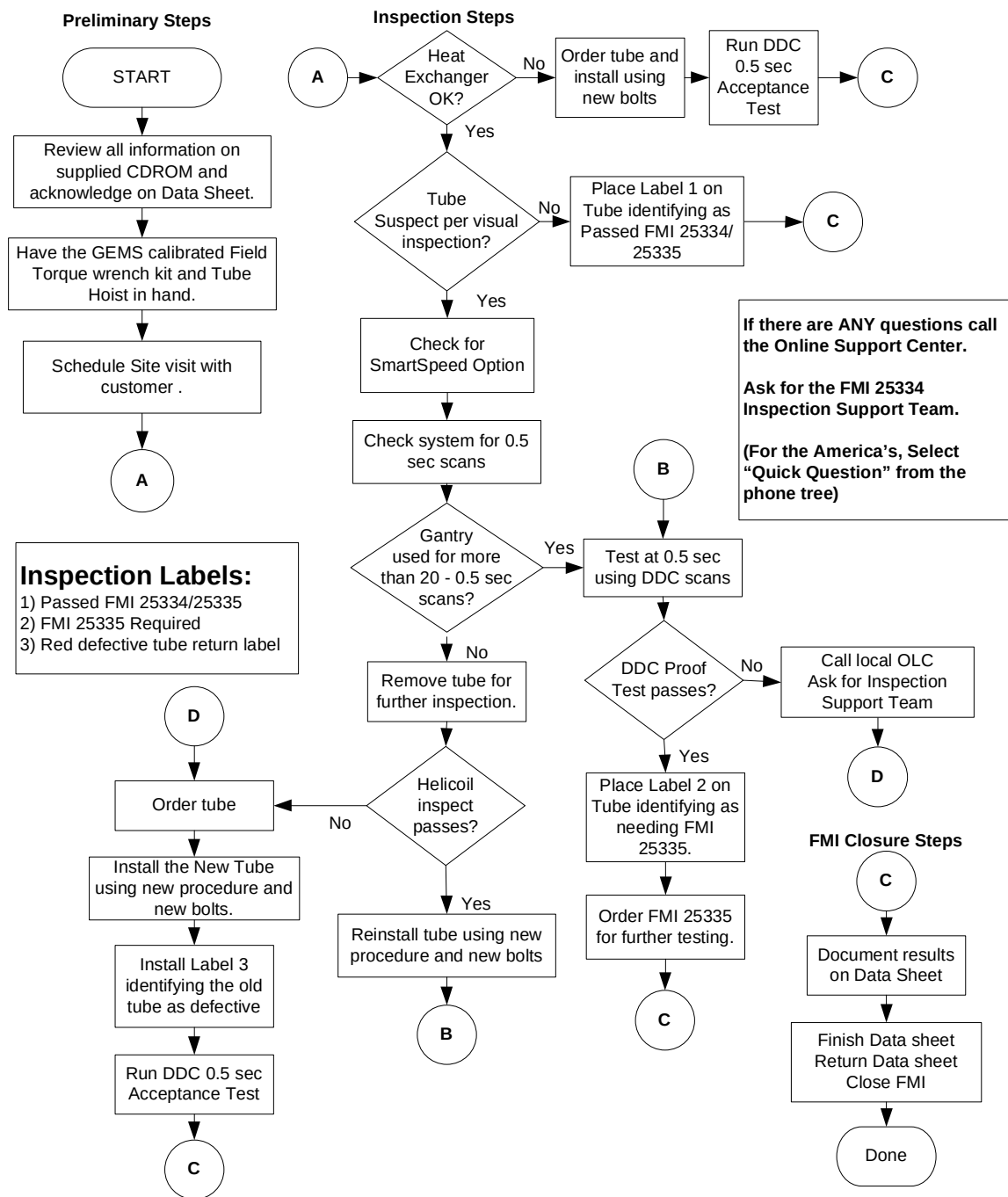
One person:

- 1.) Inspection with No Tube Removal: 2 hours labor and 1 hour of travel time.
- 2.) Tube Removal and Re-Install: 8 hours and 1 hour of travel time
- 3.) Tube Removal and Tube Replacement: 8 hours plus tube delivery time; 1 hour of travel time.

Procedure

1 Tube Casing Procedure

1.1 Performix Tube Field Inspection Process Flow



1.2 Preliminary Steps

- 1.) Review all information on the supplied CD-ROM (P/N 2399977) prior to going on site.
- 2.) There is a Data Sheet on [page 22](#) of this FMI document that must be filled out and returned upon completion of this FMI. It **MUST** be returned via two methods to guarantee receipt. An electronic copy of the data sheet is supplied on the Service Information CD-ROM (P/N 2399977) that came with this FMI. Return the data sheet via the following two methods:
 - a.) E-mail to address supplied on the data sheet (or fax, to: 414-918-1609).
 - b.) Return by regular mail, to the address supplied on the data sheet.

1.3 Heat Exchanger Check

If ANY of the inspected screws are loose or missing, you **MUST** replace the X-Ray tube. You may need a handheld expandable / retractable mirror to inspect two of the Heat Exchanger Cooler screws.

- 1.) Remove/open right side cover.



WARNING

FAILURE TO DISABLE THE AXIAL DRIVE MAY RESULT IN SERIOUS INJURY DUE TO UNEXPECTED GANTRY ROTATION.

- 2.) Disable Axial Drive via the service switch per standard service procedure for your system type.
- 3.) Remove/Open the top, side and Front Covers per standard service procedures for the system type you are currently inspecting.
- 4.) Document the Tube casing model and serial number and the date of manufacture on the data sheet at the end of this document. These can be found on the identification stickers on the Heat Exchanger.

1.3.1 Heat Exchanger Screw Inspection - Overview

During this procedure, you will visually check for the presence of 8 visible screws on the Heat Exchanger that attach the coolers.

1.3.2 Cooler Inspection (Eight Screws total)

The Heat Exchanger coolers are located on the two ends of the heat exchanger. See [Figure 1](#), below. There are eight (8) screws that mount the coolers (four (4) each), and each will be verified during this procedure.

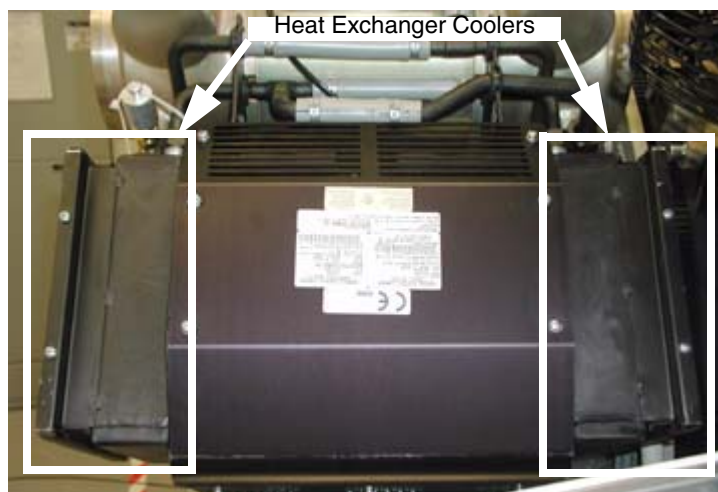


Figure 1 - Heat Exchanger Coolers

- 1.) Rotate the Gantry such that the Heat Exchanger is at the 2 o'clock position.

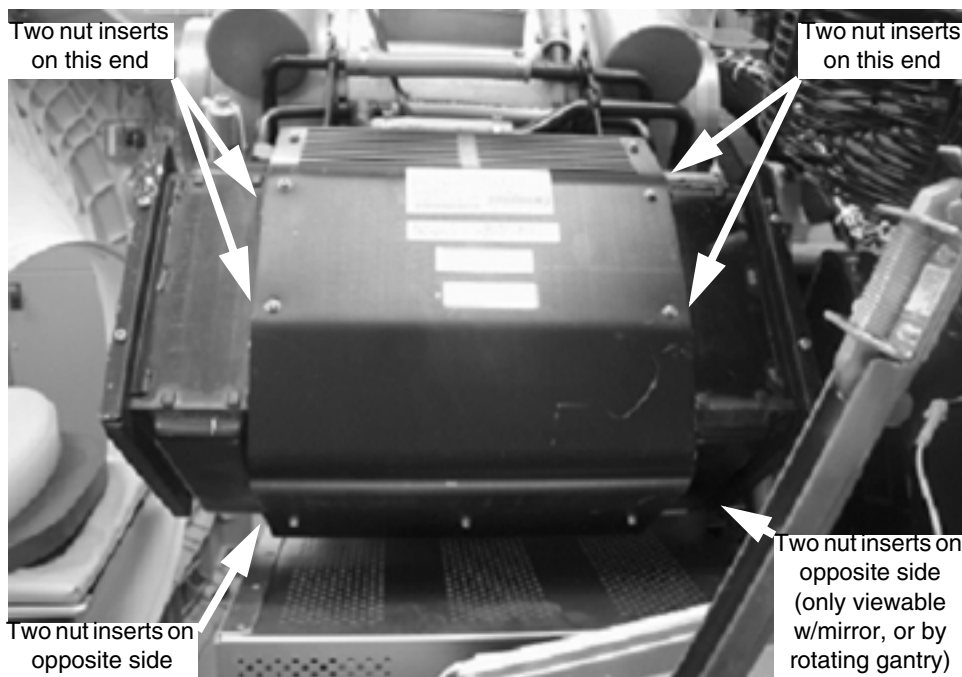


Figure 2 - Heat Exchanger at 2 o'clock

- 2.) Look at the nut inserts (shown below) on the ends of the Heat Exchanger to verify that the end of the internal screws are seen at the end of the nut inserts as seen in [Figure 3](#). If any of the screws are missing, replace Tube:
 - a.) Go to [Section 2](#), to order a new Tube,
 - b.) Go to the [Tube Replacement](#) Section, for installation instructions, and
 - c.) Proceed to [FMI Completion](#).

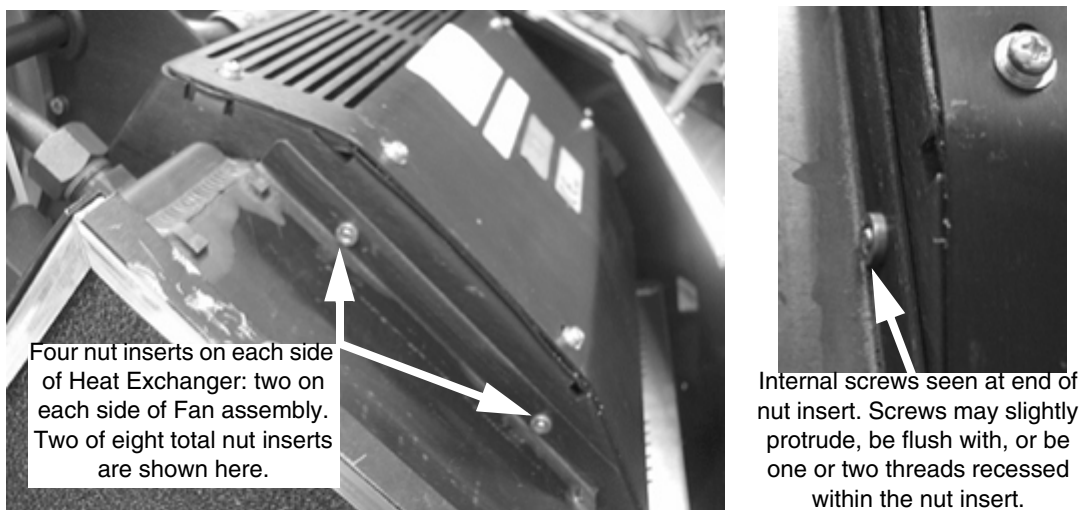


Figure 3 - Two of Eight nut inserts shown (left); screw seen just at end of nut insert (rt)

- 3.) With the Heat Exchanger at 12 o'clock, you can see the bottom nut insert, by looking from either side of gantry. Using a mirror and/or rotating the gantry into multiple locations you can view the inside cooler screws that are nearest the slipping per Figures below. If any of the screws are missing or loose, replace Tube:
- Go to [Section 2](#), to order a new Tube,
 - Go to the [Tube Replacement](#) Section, for installation instructions, and
 - Proceed to [FMI Completion](#).

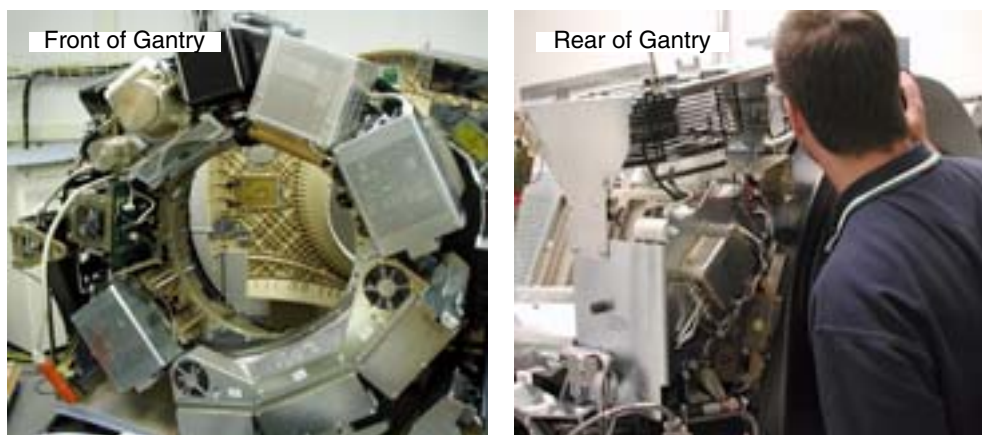


Figure 4 - Gantry position to verify 1 of the 2 “difficult to see” cooler screws, and where to look to verify screw (LightSpeed Ultra System Shown)

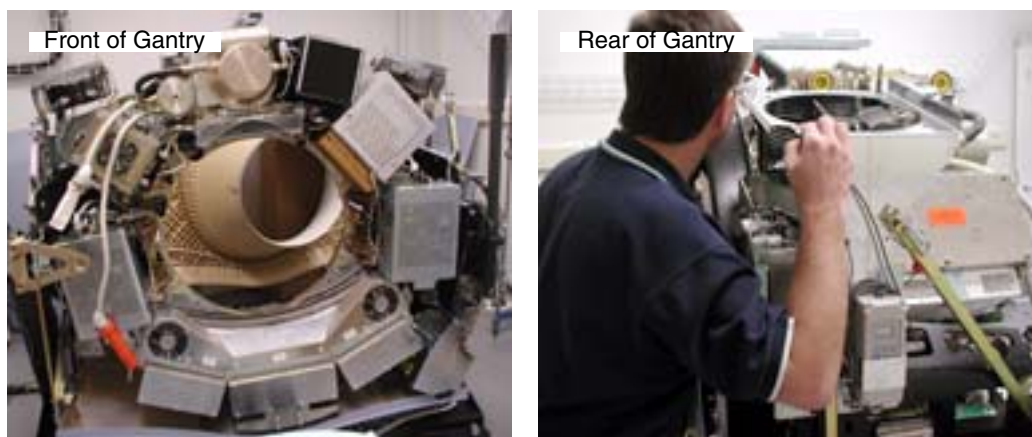


Figure 5 - Gantry position to verify the other 1 of the 2 “difficult to see” cooler screws, and where to look to verify screw (LightSpeed Ultra System Shown)

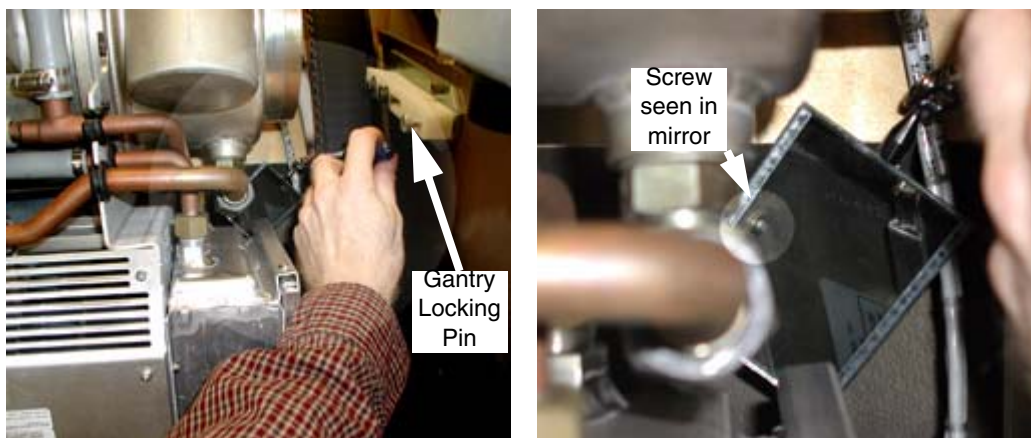


Figure 6 - Using a mirror to view “difficult to see” cooler screws (LightSpeed QX/i gantry shown)

- 4.) Verify that each cooler is secure by attempting to move in three ways: rotation, up/down/left/right, and in/out per [Figure 7](#), below. Do NOT forcefully attempt to move the cooler.

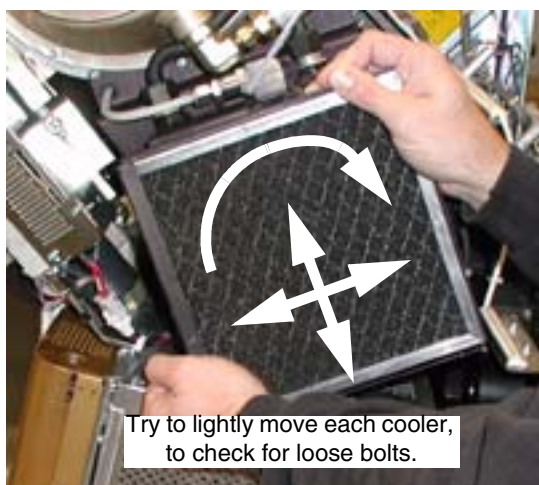


Figure 7 - Example bolt check method

If loose, replace the tube:

- Go to [Section 2](#), to order a new Tube,
- Go to the [Tube Replacement](#) Section, for installation instructions, and
- Proceed to [FMI Completion](#).

If this step *passes*, continue with the next section.

1.4 Tube Visual Check - On Gantry

At the end of this document is a data sheet that you **MUST** fill out and return to the address on the data sheet when this FMI is completed. First item to fill in is the System Identification number for the system on which you are performing this FMI.

- 1.) Rotate the Gantry to have the tube at the 3 o'clock position.



Figure 8 - Tube at 3 o'clock position - Pointing to location of interest

- 2.) Look at the tube casing welds. Use the pictures below to help identify a suspect tube. Use the Video clip titled "Performix_Tube_FMI25334_Video.mpg" on the supplied CD-ROM (P/N 2399977) to help identify the method of looking at the areas shown above. Also available on the CD-ROM are the color versions of the pictures provided here.

Note: If you have identified a suspect tube, call the Online Support Center to notify them that a suspect tube has been found. For the Americas, select the "Quick Question" option from the phone tree.

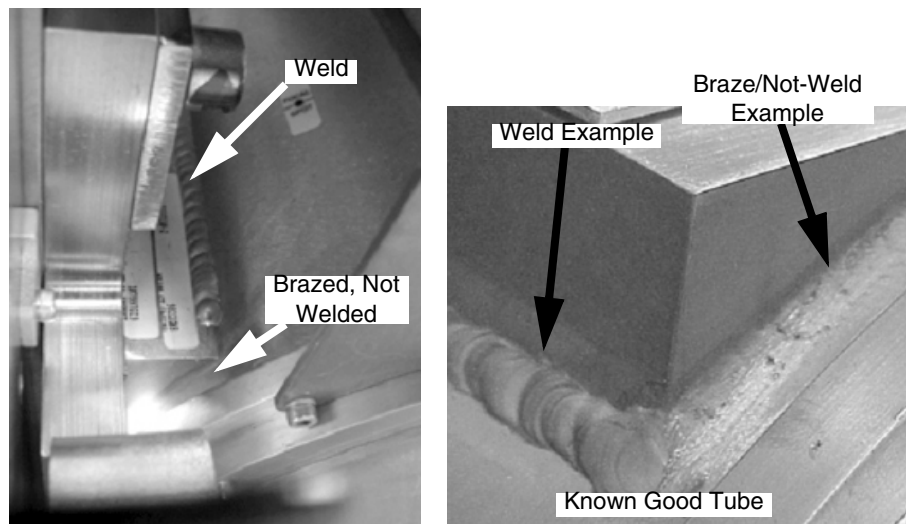


Figure 9 - Weld on one side; Close up of Braze/Not-Weld and Weld (right) - Inspection Passes, No Problem with this tube casing. Go to ["FMI Completion."](#)

The photo at the right of [Figure 9](#) shows an example of a Braze that does not look smooth, but is still different from a weld. This is intended to improve proper identification of a weld.

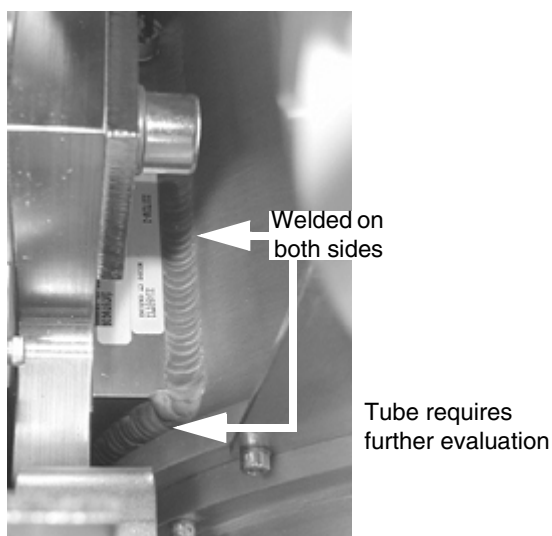


Figure 10 - Weld seen on both sides

- 3.) Depending on the weld configuration seen, perform one of the following actions:
 - a.) If the welds are seen on both sides, as shown in [Figure 10](#), then your tube casing is suspect. You must continue with [Section 1.5 - SmartSpeed Option Check](#).
 - b.) If the welds are *not* on all sides, as shown in [Figure 9](#), then:
 - * Install label 1, "Passed FMI 25334/25335" (P/N 2400473), identifying the tube as inspected.
 - * Finish the data sheet on [page 22](#), and return it per instructions.
 - * Go to the [FMI Completion](#) section.

1.5 SmartSpeed Option Check

This section will check the system options to see if the Smartspeed option has been installed.

- 1.) Open a C-shell via the Common Service Desktop Utilities Menu - Utilities Folder - Shell.
- 2.) Type the following command:
`swokininstall -p`
- 3.) Check the list of installed options and look for the "Smart Speed" option, as shown below:
Product Software Option Keys:

Smart Speed Permanent
- 4.) Document this on the data sheet on [page 22](#).

1.6 Customer Scan Speed Check

This section will check the system to see if the system has been run at 0.5 second rotation speeds. If the system has been run at 0.5 second gantry rotational speeds then only a verification test is required. If the system has NOT been run at 0.5 second gantry rotational speeds, then another visual inspection of the mounting hardware is required. This extra check is explained in [Section 1.7](#). There are two methods of performing this check.

1.6.1 Method 1

- 1.) Open a C-shell via the Common Service Desktop Utilities Menu - Utilities Folder - Shell.
- 2.) Type the following commands, make sure to keep the spaces as shown:

```
cd /usr/g/bin
PrintScanUsage -d | awk -F: 'BEGIN{c=0} {if ($25=="0.5") {c++}} END{print c}'
```

spaces: ^ ^ ^ ^ ^ ^ ^ ^ ^ ^
- 3.) You will get a number for a result, indicating the number of 0.5 second scans run.
 - a.) If less than 20 “0.5” second scans are seen, continue with [Section 1.7 - Inspections - Off Gantry](#).
 - b.) If 20 or more “0.5” second scans are seen, continue with [Section 1.6.3](#).

1.6.2 Method 2

- 1.) Open the Common Service Desktop from the service window of the Operator Console
- 2.) Under the “Error Logs” tab, select the “System Browser (all features)” utility.
- 3.) Click on the “Log Select” button at the top of the browser, and, from the pull-down list, select “Scan Usage”. Once selected, click on the “Show Log!” button.
- 4.) The browser window displays a list of up to the last 33,000 scans, starting with the last scan taken. The list is shown 200 lines at a time. Click on “Next” (at the bottom of the list) to see the next 200 lines. Scroll to the top of the list to see the display's column naming convention. Scroll to the right looking for the “gantry_period_sec_rev” column - this the gantry rotation per second field.
- 5.) Scroll down the list while watching this column to see whether there is a “0.5” second rotation selection.
- 6.) Continue to scroll and count the number of “0.5” second scans.
 - a.) If less than 20 “0.5” second scans are seen, continue with [Section 1.7 - Inspections - Off Gantry](#).
 - b.) If 20 or more “0.5” second scans are seen, continue with the next step in this section.

1.6.3 Verification Scan

If the customer has been using the system at 0.5 second gantry rotation speeds per the check in step 6, a verification test using a Diagnostic Data Collection (DDC) scan needs to be run.

Perform the following steps:

 **WARNING** **PRIOR TO (AND DURING) ANY SCAN, VERIFY THAT NO PERSONNEL ARE IN THE SAME ROOM/AREA AS THE GANTRY. DO NOT GO IN THE SCAN ROOM DURING THIS TEST.**

- 1.) Setup for the CBF alignment check as follows:
 - a.) Select SERVICE DESKTOP.
 - b.) Select REPLACEMENT
 - c.) Select TUBE CHANGE

- d.) Select CBF AND SAG ALIGNMENT
- 2.) Run the scans and calculate the results as stated by the on screen instructions.
- 3.) Document the resulting adjustment shown by the program in [Table 1](#). DO NOT make any adjustments. This is for a check only.
- 4.) Set for the ISO alignment check as follows:
 - a.) Select SERVICE DESKTOP
 - b.) Select REPLACEMENT
 - c.) Select TUBE CHANGE
 - d.) Select ISO ALIGNMENT
- 5.) Run the scans and calculate the results as stated by the on screen instructions.
- 6.) Document the resulting adjustment shown by the program in [Table 2](#). DO NOT make any adjustments.
- 7.) Move the pin used for the CBF and ISO scans out of the scan plane.

**WARNING****DO NOT GO NEAR THE GANTRY DURING THIS TEST.**

- 8.) Setup a DDC scan as shown in [Appendix A - Using DDC](#), and run this scan five (5) times. This will run the gantry at 0.5 second rotations for 10 minutes.
- 9.) Run the CBF and ISO alignment checks again just as in steps [1-6](#) above.
- 10.) Document the results for both below:
- 11.) Calculate the difference in adjustments for the “inch” results only in [Table 2](#).

Table 1 - Adjustment results from CBF and ISO calibration checks

	CBF adjustment in mm	CBF adjustment in inches	ISO adjustment in mm	ISO adjustment in inches
Prior to DDC scans				
After DDC scans				
Difference	N/A		N/A	

- 12.) Perform a visual check of the tube.
 - a.) Position the tube at the 4 o'clock position (toward the bottom of the gantry).
 - b.) Take a piece of paper and try to insert it between the tube casing plate and the tube mounting plate. Do NOT check between the mounting plate and collimator. See [Figure 11](#) below.

Do NOT check
between the
mounting plate
and collimator.

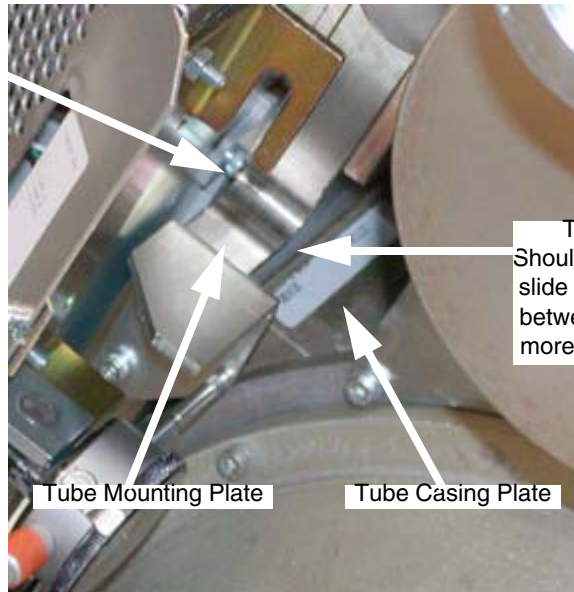


Figure 11 - Tube casing and Mounting plates

- c.) If the paper can be inserted more than $\frac{1}{4}$ " (6mm) in between the tube casing plate and the tube mounting plate, call the Online Support Center and ask for the Inspection Support Team.
- 13.) Depending on the results of this check perform one of the following actions:
- If the CBF or ISO difference is greater than or equal to 0.003 inches then:
 - a.) Contact the Online Center and ask to be conferenced with the Tube Inspection Support Team.
 - b.) Follow the actions as defined by the Tube Inspection Support Team.
 - c.) Document the results on the data sheet on page 20, and return it per instructions.
 - d.) Go to the [FMI Completion](#) section.
 - If the CBF or ISO difference is less than 0.003 inches then:
 - a.) Install label 2, "FMI25335 Required" (P/N 2401457), identifying the tube as requiring FMI 25335.
 - b.) Document the results on the data sheet on [page 22](#), and return it per instructions.
 - c.) Order FMI 25335 for further testing. Wait until receiving FMI 25335 to schedule another site visit. Go to the [FMI Completion](#) section.

1.7 Inspections - Off Gantry

Important: At this time call the Online Support Center for help. Ask for assistance from the FMI25334 Inspection Support Team. (For the Americas, select the "Quick Question" option from the phone tree.) Tell them the results of the inspections up to this point and that you are taking a tube off for Helicoil inspection. Call *before* removing the tube.

1.7.1 Visual Inspection - Off Gantry



WARNING

USE ONLY THE BOLT KIT SPECIFIC TO YOUR SYSTEM WHEN INSTALLING THE NEW TUBE OR RE-INSTALLING CURRENT TUBE. DO NOT REUSE BOLTS.

**CAUTION**

Tube and tube oil may be hot, depending on recent scan load. Wait 30 minutes after last scan, with power to the gantry, to allow the tube and tube oil to cool, prior to removal.

- 1.) Use standard service procedures specific to your system type for removing the tube. Remove Mounting Plate and discard mounting hardware.
- 2.) Confirm previous visual inspection with tube off gantry. Use the following pictures to help identify a suspect tube. If you are not sure, call the Online Support Center for help. Ask for assistance from the FMI25334 Inspection Support Team. (For the Americas, select the "Quick Question" option from the phone tree.) If you can not contact the support team, consider the tube as suspect and continue with [Section 1.7.2](#).

If the visual inspection with the tube off the gantry shows the tube as NOT suspect, per previous definitions and figures, then:

- a.) Install label 1 "Passed FMI 25334/25335" (Part # 2400473), identifying the tube as inspected.
- b.) Finish the data sheet on [page 22](#) and return it per instructions.
- c.) Go to the [FMI Completion](#) section.

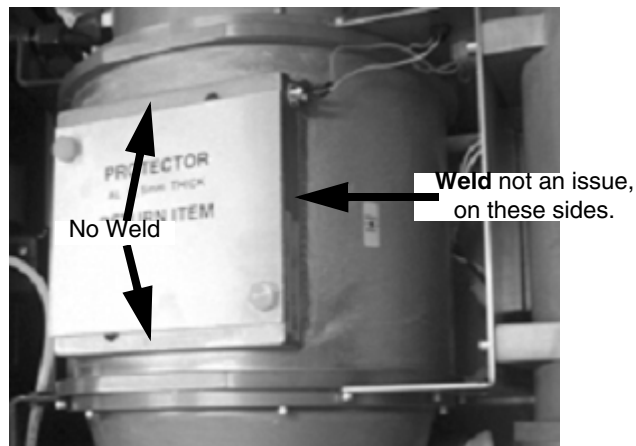


Figure 12 - Tube not suspect

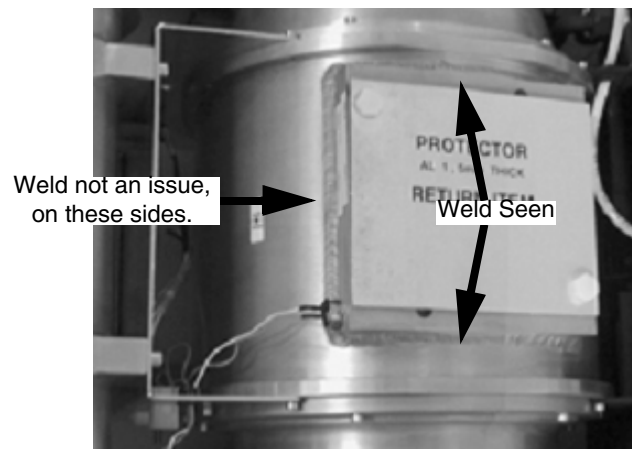


Figure 13 - Suspect tube (requires further inspection)

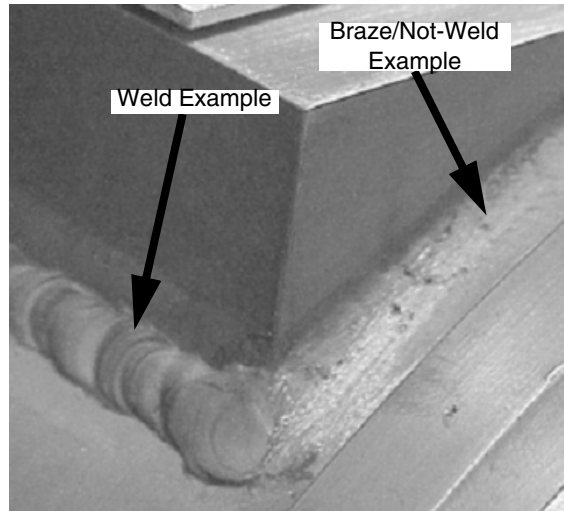


Figure 14 - Close up of Braze/Not-Weld and Weld

Figure 14 shows an example of a Braze that does not look smooth, but is still different from a weld. This is intended to improve proper identification of a weld.

1.7.2 Helicoil Visual Inspection

Perform a visual inspection of the helicoils in the tube case mounting holes as identified by Figure 15, using the process defined in this section. Use a flashlight for good lighting and a magnifying glass, if available. A helicoil is a threaded insert used in a threaded mounting hole, for increased mounting strength.

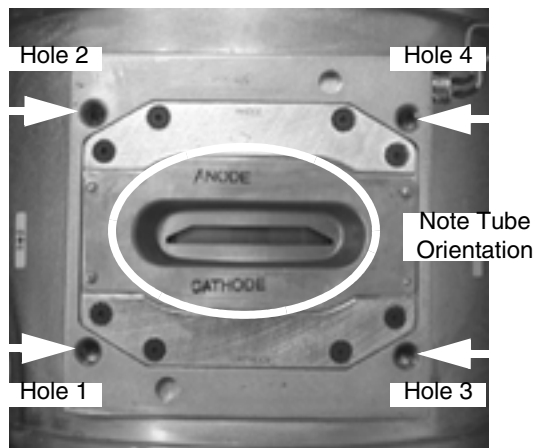


Figure 15 - Mounting Holes for Inspection

The helicoils used in this tube are “locking helicoils.” This means that they have a squared section in the middle as shown in Figure 16, below.

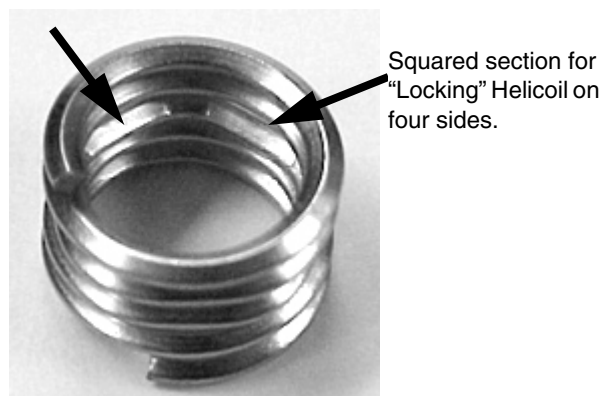


Figure 16 - Helicoil Insert Outside Tube

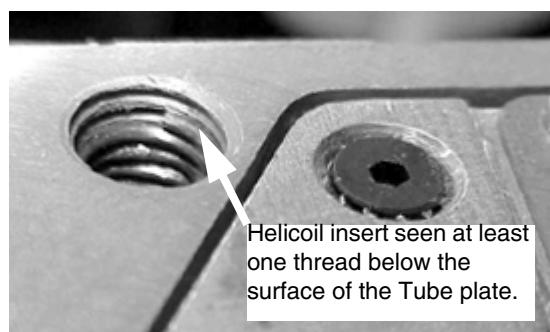


Figure 17 - Helicoil in Mounting Hole

- 1.) Using the pictures above, and the drawing below, make sure the helicoil is recessed at least one thread below the surface of the tube casing plate.
- 2.) Anything large enough to be seen as a metal shaving or sliver (greater than 2mm long) is NOT normal and requires a tube replacement.

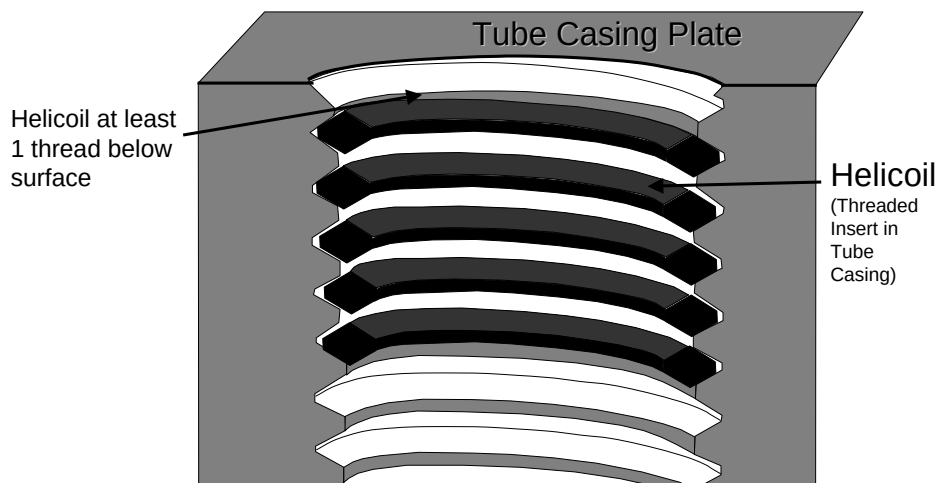


Figure 18 - Cross-section of tapped hole with Helicoil Insert

-  **WARNING** **CLEANING/REMOVING METAL SHAVINGS OR SLIVERS, AS DEFINED IN STEP 2, WILL NOT REMOVE THE PROBLEM. THE TUBE MUST BE REPLACED, IN THIS INSTANCE.**
- 3.) Perform one of the two following actions (as is appropriate):
- a.) If the helicoil inspection as defined in this section *fails*, the tube must be replaced. Fill in the "Helicoil" inspection results in the data sheet on [page 22](#), and go to [Section 2](#) for Tube replacement.
 - b.) If the helicoil inspection *passes*, go to [Section 2.2](#), for Installing the tube using new instructions not available on the current revision of the Service Methods CD-ROM. Install label 2, "FMI25335 Required" (P/N 2401457), identifying the tube as requiring FMI 25335.

2 Tube Replacement

2.1 Order New Tube

Replace the tube, if the cooler or helicoil inspections fail. You will need to inform the customer that the tube has to be replaced, along with estimated time for tube arrival. Also inform the local sales and service managers that the customer will be down for a tube change.

QTY	PART #	DESCRIPTION	USED ON
1	D3182T	Performix Ultra Tube:	LightSpeed 2X, 3X, 4X, and Discovery LS, DS

-  **NOTICE** **If you are on a government site that requires new replacement parts instead of refurbished parts use the following table to order the proper tube.**

QTY	PART #	DESCRIPTION	USED ON
1	D3181T	Performix Ultra Tube:	LightSpeed 2X, 3X, 4X, and Discovery LS, DS

2.2 Installing Tube

2.2.1 Procedure

-  **WARNING** **ONLY USE THE TORQUE WRENCH TO PUT IN TUBE BOLTS. THIS IS A LOW TORQUE BOLT. USE OF ANYTHING OTHER THAN A TORQUE WRENCH MAY RESULT IN THREAD DAMAGE.**

-  **NOTICE** **REQUIRED: View the video titled "Proper use of a Torque wrench" on the CD-ROM (P/N 2399977) supplied with this FMI prior to installing the tube mounting plate and tube.**

- 1.) Use NEW Bolt Kits, provided with the replacement tube, to attach:
 - a.) Mounting Plate to the Tube
 - b.) Tube to the Gantry

-  **CAUTION** **ONLY use the bolt kit labeled for the system upon which the tube is used.**
- 2.) Install the tube per instructions provided in [Appendix D - Tube Replacement: HiSpeed QX/i, LightSpeed, Discovery LS/ST](#).

Note: Do NOT use the Tube Replacement procedures found on the current service CD release.

**NOTICE**

The tube must be allowed to cool for five (5) hours prior to running calibrations. Check to make certain that five hours have passed since the last scan. Wait until the five hours have passed, before proceeding.

- 3.) Align and calibrate your system per standard service procedures for your system type.

2.2.2 Verification Scan

Perform the following steps:

**WARNING****DO NOT GO NEAR THE GANTRY DURING THIS TEST.**

- 1.) Setup for the CBF alignment check as follows:
 - a.) Select SERVICE DESKTOP.
 - b.) Select REPLACEMENT
 - c.) Select TUBE CHANGE
 - d.) Select CBF AND SAG ALIGNMENT
- 2.) Run the scans and calculate the results as stated by the on screen instructions.
- 3.) Document the resulting adjustment shown by the program in [Table 2](#). DO NOT make any adjustments.
- 4.) Set for the ISO alignment check as follows:
 - a.) Select SERVICE DESKTOP
 - b.) Select REPLACEMENT
 - c.) Select TUBE CHANGE
 - d.) Select ISO ALIGNMENT
- 5.) Run the scans and calculate the results as stated by the on screen instructions.
- 6.) Document the resulting adjustment shown by the program in [Table 2](#). DO NOT make any adjustments.
- 7.) Move the pin used for the CBF and ISO scans out of the scan plane.
- 8.) Setup a DDC scan as shown in [Appendix A - Using DDC](#), and run this scan five (5) times. This will run the gantry at 0.5 second rotations for 10 minutes.
- 9.) Run the CBF and ISO alignment checks again just as in steps 1-6 above.
- 10.) Document the results for both below:
- 11.) Calculate the difference in adjustments for the “inch” results only in [Table 2](#).

Table 2 - Adjustment results from CBF and ISO calibration checks

	CBF adjustment in mm	CBF adjustment in inches	ISO adjustment in mm	ISO adjustment in inches
Prior to DDC scans				
After DDC scans				
Difference	N/A		N/A	

- 12.) Perform a visual check of the tube.
- Position the tube at the 4 o'clock position (bottom of the gantry).
 - Take a piece of paper and try to insert it between the tube casing plate and the tube mounting plate. Do NOT check between the mounting plate and collimator. See [Figure 19](#) below.

Do NOT check
between the
mounting plate
and collimator.

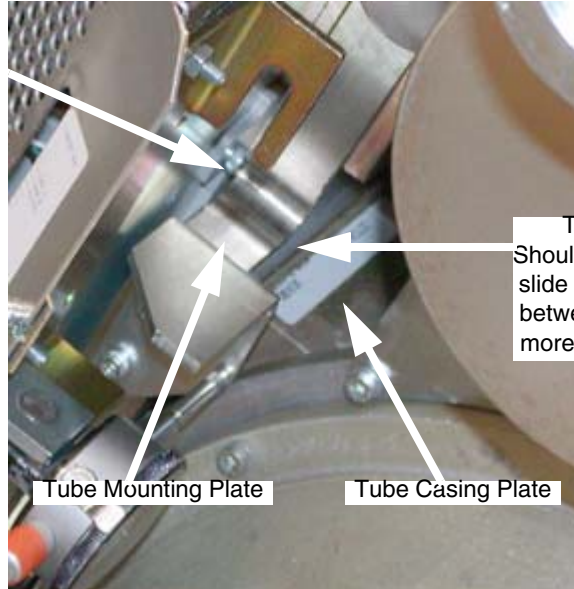


Figure 19 - Tube casing and Mounting plates

- If the paper can be inserted more than $\frac{1}{4}$ " (6mm) in between the tube casing plate and the tube mounting plate, call the Online Support Center and ask for the Inspection Support Team.
- 13.) Depending on the results of this check perform one of the following actions:
- If the CBF or ISO difference is greater than or equal to 0.003 inches then:
 - Contact the Online Center and ask to be conferenced with the Tube Inspection Support Team.
 - Follow the actions as defined by the Tube Inspection Support Team.
 - Document the results on the data sheet on page 20, and return it per instructions.
 - Go to the [FMI Completion](#) section.
 - If the CBF or ISO difference is less than 0.003 inches then:
 - Install label 2, "FMI25335 Required" (P/N 2401457), identifying the tube as requiring FMI 25335.
 - Document the results on the data sheet on [page 22](#), and return it per instructions.
 - Order FMI 25335 for further testing. Wait until receiving FMI 25335 to schedule another site visit. Go to the [FMI Completion](#) section.

2.3 Returning Tube

Attach one of the supplied Red “Defective Tube Hardness Return” stickers (P/N 2358146) on the tube, in the location shown in [Figure 20](#), below. Place the other defective tube sticker on the tube crate. Return the tube by the standard tube return process.

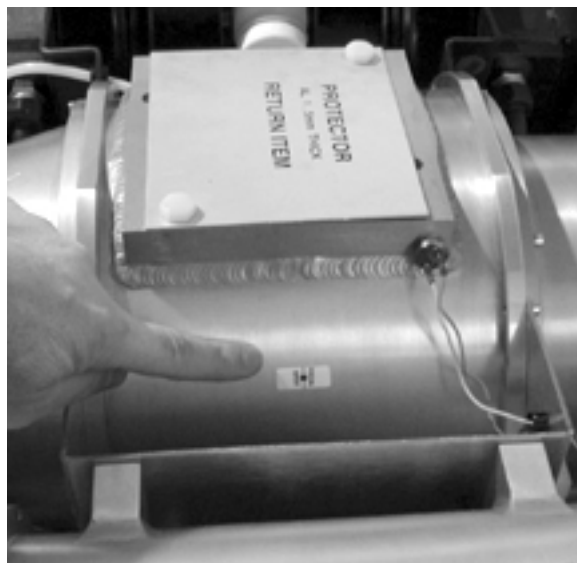


Figure 20 - “Tube Defective” Sticker Location

FMI Completion

The table below shows a summary of possible results of this FMI to help clarify if the more detailed inspections defined in FMI 25335 are required at a later date.

Table 3 - Summary of FMI25334 Results and Actions

FMI Result	Action
If the Heat Exchanger or Helicoil inspection fails:	Order a new tube and Close this FMI per this section. FMI 25335 is not necessary.
If the tube is identified as requiring further inspection (has welds on all sides of tube casing plate) and is not being replaced at this time:	Order FMI 25335 and Close this FMI per this section.
If the inspections have passed:	Close this FMI per this section, FMI 25335 is not necessary.



NOTICE **Data sheet must be returned via e-mail, within 24 hours.**

- 1.) Complete the data sheet supplied electronically on the CD-ROM (P/N 2399977), and e-mail it per instructions on the data sheet, within 24 hours. Also send a hard copy via regular mail, within three (3) days.
- 2.) Update Global Installed Base (GIB) database on the web. Remove the tube casing serial numbers and tube insert serial numbers that are no longer on this CT System. Go to [Appendix B - Global Installed Base \(GIB\) Update Instructions](#), and follow the process specific to your region.

- 3.) If a Tube is replaced:
 - a.) Make sure the new tube has an “Inspected” sticker, as shown in [Figure 21](#), below. If not, return to [Section 1.7](#) of this document to verify visual inspection of the replacement tube, prior to installing the new tube.



Figure 21 - “Tube Inspected” Sticker Location

- b.) Load the Product Locator Card information into GIB (see [step 2](#)).
- 4.) Remove the FMI Document from the customer site.
- 5.) If any section of this inspection process has identified the tube as requiring further inspection, then order FMI25335. Make certain that the data sheet from this FMI ([page 22](#)) is returned via e-mail *and* regular mail, per the instructions on the data sheet. The subject line should read either:
 - “FMI 25335 Needed,” if ordering FMI25335, *or*
 - “FMI25334,” for all other instances.

FMI 25334 Data Sheet

PERFORMIX INSPECTION CHECKLIST

Complete this Data sheet. E-mail a copy (or fax, to: 414-918-1609), within 24 hours of FMI completion, to: **Performix.Inspection@med.ge.com**, with the subject line "FMI 25335 Needed," if ordering FMI25335 — otherwise, the subject line should read "FMI 25334." Send the signed copy via regular mail, within three (3) days, to:

Andrew Rubin, EA36
General Electric Company
4855 W. Electric Avenue
Milwaukee, WI 53219

Initial the boxes below to confirm that you have reviewed the supplied videos prior to performing this FMI. Review all information on the supplied CD-ROM (P/N 2399977).

	Torque Wrench Video	Performix Tube FMI Video	Heat Exchanger Video
CD-ROM Videos Reviewed			

Complete the tables below for the system being inspected:

Torque Wrench Tracking Number	
-------------------------------	--

Hospital Name			
GEMS System ID			
System Type (Circle)	LightSpeed 2.x, 3.x, 4.x, HiSpeed QX/i, Discovery LS/ST		
SmartSpeed option installed	YES / NO	Customer runs 0.5 sec scans	YES / NO
Gantry Model Number			
Gantry Serial Number			
Insert Serial Number			
Casing Serial Number			
Date of Manufacture			
Inspector / Date			

All inspections may not have been done. Document only those inspections that were actually performed. Circle the answer that applies:

Heat Exchanger Cooler Screw Inspection

8 screws installed	Yes / No
Coolers solidly mounted	Yes / No

Tube Casing **Weld** Inspection Results

4 Welds, Suspect Tube	Yes / No
-----------------------	----------

	Location/Hole 1	Location/Hole 2	Location/Hole 3	Location/Hole 4
Helicoil Inspection Results	Pass / Fail	Pass / Fail	Pass / Fail	Pass / Fail

Tube changed at time of this inspection:	YES / NO
--	----------

IF ANY TUBE FAILS THE HEAT EXCHANGER OR HELICOIL INSPECTIONS, IT MUST BE REPLACED WITH A NEW TUBE, NOW. THE TUBE THAT FAILED INSPECTION IS RETURNED PER NORMAL PROCEDURES, WITH THE SUPPLIED LABEL ATTACHED.

Appendix A - Using DDC

Using DDC to Rotate Gantry for 2 minutes at fastest allowed Gantry Speed

1.

Launch Diagnostic Data Collection (DDC) Utility:

- Open the Common Service Desktop
- Click on the Diagnostics Icon
- Click on the Data Acquisition Folder
- Click on the Diagnostic Data Collection Menu Item

2.

Prepare System For DDC Scan:

- Select **Rotating X-Ray Off**
- Type **120** in Scan Time (sec) field
- Type **0.5** (or fastest allowed speed for your system) in the Gantry Velocity (sec/rev) field (verify other settings as shown)
- Click on **Accept Rx** button to send scan request to firmware

The Utility:

Displays the Scan List GUI and the Real Time Information window.

After a few seconds, starts flashing the **Scan** button on the console

3.

Press the Scan button on the Console to start the DDC Scan. When complete, the DDC Utility dismisses the Scan List GUI and the Real Time Information window.

Protocol Name: ddc_axial_xray_off

Scan Type: Static X-Ray Off, Static X-Ray On, **Rotating X-Ray Off**, Rotating X-Ray On

Miscellany: Protocol Name, Position Tube, Show Request, Monitor Enable

Run Description: Scan Time (sec) 120.00, Gantry Velocity (sec/rev) 0.50, No. of Scans 1, Scan Start Position (degrees) 0.00, InterScan Delay (sec) 2.00, X-Ray Duration, kV 80, Dry Unit X-Ray On (0 - 29.9)(sec), mA 0, Gantry Tilt (Degree) 0.0, DAS Gain [1-31] 3, Rotor RxOverrides On Off, Trigger Rate (Hz) 984 1090 1230 1312 1400 **1690**, Filter Head **Body** Blocked Air, Focal Spot Small Large, Slice Collimation 4x1.25 **4x2.50** 4x3.75 4x5.00, FET Mode Overrides Off, Collimator Width None Full Non-Tracking Scan Non-Tracking Cal, Options Auto Scan **TOX1**

Buttons: Save, **Accept Rx**, Dismiss

Real Time Information

Patient Handling

Scanning

Scanner Hardware Stopped Scan

Delay Timer

Scan List

kV	mA	Thickness	Filter	Scan Time	Spot Size	Scan Number
80	0	4x1.250	BODY	120.00	SMALL	1

Buttons: Cancel, Suite: CTS3, Exam: 50964, Series: 1, Total Scans: 1, Remaining Scans: 1, Resume

Appendix B - Global Installed Base (GIB) Update Instructions

1 United States Instructions



NOTICE The instructions in this section are for the Americas Pole only. Instructions for the Europe Pole may be found in [Section 2 \(on page 28\)](#). Likewise, instructions for the Asia Pole may be found in [Section 3 \(on page 28\)](#).

These instructions take you through the steps required to login to the GIB system via the web and update an installed CT system for Tube Casing and Tube Insert Serial Numbers.

Use the link below to access the GIB web site.

http://egems.gemedicalsystems.com/gib/gib_MainMenu.jsp

- 1.) The first screen you will see is the login screen. Log in using your SSO and password.

GEMS Login!

Please enter your GEMS SSO User ID and Password.

User Name:

Password:

Login

- 2.) After logging in, the next screen you will see is shown below. Select "Global Installed Base entries".

- 3.) The next screen shown below is the Main entry screen. Enter your GE Cares System ID in the field provided at the bottom of the screen and select "Submit".

Home
Global Installed Base Entry

Installation Process

Order nbr : (Example : xxxxxxxx)

OR

OCP : (Example : xxxxx)

FDO : (Example : xxxxxxxx)

B/S : (Example : xxx)

Country(ISO): US

Serial Changes to Installed Base

For current installed base update to GIB,
enter system id that requires
modifications and click 'Submit'.

System ID: (Example : xxxxxxxxxxxxxxxxx)

Country(ISO): US

- 4.) This will pull up a listing of Serial numbers attached to the CT system ID. Below is an example of the CT Casing and Insert Serial Numbers in the format you will see. The Line # shown is purely for list purposes and just depends on the order the serial numbers were originally entered for this system.

Line # : 007

Model: 2137130-2

Serial: 00000018944TY9

Desc.: MX200 CT CASING

Inst./Deinst Date (YYYY-MM-DD):

Delete (D):

Reason Code:

Line #: 012

Model: 2120785

Serial: 00000106590GI3

Desc.: MX200 CT INSERT

Inst./Deinst Date (YYYY-MM-DD): 2003-11-26

Delete (D):

Reason Code:

- 5.) To delete Serial numbers for Tube Casings and Inserts that no longer exist on this system, enter a "D" in the Delete field shown in the above examples. When you have done this for any items on the current displayed page, select "Submit" to update GIB.
- 6.) The screen below shows the "Submit" button and in this instance that there are 3 pages of serial numbers attached to this system.

[Home](#) **GIB - Changes to Installed Base** 

[Submit](#)

SystemId: 309647GCT1

Name: GRAHAM HOSPITAL ASSOC/AMHS

System Install Date: 2000-07-02

OCP:

FDO:

BS:

Submit changes before
going to another page.

Pages >>> [1](#) [2](#) [3](#) [New Entries](#)

2 Europe Instructions

For the system being inspected, document the following information and send it by e-mail to the address listed below.

- System country name:
- Gantry Model Number:
- Gantry Serial Number:
- Tube Case Model Number:
- Tube Case Serial Number:
- Tube Insert Model Number:
- Tube Insert Serial Number:
- MUST System ID:

Send this information to: **GIB, European Central Admin (MED)**

Please note in the e-mail to have any other Tube or Insert entries not listed above to be removed from GIB for this system.

3 Canada/Latin America/Asia Instructions

Follow local procedure to update the Tube Case and Tube Insert model and serial numbers for the CT system being inspected.

Remove any Tube or Insert entries in GIB that do not match the tube currently on the system.

Appendix C - Tube Bolt Information

1 Overview

This appendix contains the message text of a notice that has previously been sent to the full Field Service distribution.

THIS MESSAGE HAS GONE OUT TO CT GENERAL DISTRIBUTION.

2 Tube Bolt Message

2.1 Equipment

LightSpeed 1X, 2X, 3X, 4X and HiSpeed QX/i

2.2 Updates

- 1.) This week another LightSpeed system was identified as having 2 wrong bolts (too long) installed during a recent tube change. After the tube change an IQ issue was seen and then the FE proactively removed the tube and verified that 2 CT/i bolts had been installed during the June 24th tube change. When installing a replacement tube make sure the Mounting Plate is attached with four M10 x 25mm bolts.
- 2.) When performing a Tube Change replace the tube/mounting plate bolts with the four M10 x 25mm provided in the hardware kit. Measure the length of the bolts and verify prior to installing. The service documentation will be updated to add "replace the bolts". Also measure the length of the "old" bolts, if any of them are 30mm please generate a Product Quality Report (PQR) http://itrak.med.ge.com/gemsiTrak/ExtraView?p_action=getMain&p_option=security.LoginDisplay and save the hardware for evaluation by engineering.
- 3.) If you experience motion IQ issues (aliasing) after a LightSpeed Performix Tube Change (up to 120 Days after tube change) verify that the LightSpeed mounting bolts were used.
- 4.) If you have any questions on tube mounting hardware call your OLC for clarification prior to mounting the tube.
- 5.) To eliminate the mixing of bolts the existing Performix Tube hardware kit will be split into product specific packages. You should see the new hardware kits by middle of August.
Performix ADV Tube D3172T: LightSpeed H/W Kit (P/N 2391264-2) + CT/i H/W Kit (P/N 2391264)
Performix Ultra Tube D3182T: LightSpeed H/W Kit (P/N 2391264-2)
- 6.) Manufacturing is inspecting all new systems to ensure correct bolts are in forward production systems.
- 7.) Engineering is evaluating IQ tests that would identify systems with tubes mounted with incorrect hardware.
- 8.) There will be follow-up communications in the near future.

2.3 Problem

The Performix Tube is shipped with a mounting hardware kit that contains different length bolts for use on CT/i, LightSpeed/HiSpeed QX/i, and HiSpeed ZX/i and HiSpeed NX/i products. CT Manufacturing mounts the Tube with the four M10 x 25mm bolts from the hardware kit. For field tube replacements the LightSpeed 1X, 2X, 3X, 4X and HiSpeed QX/i Service procedure specify to reuse the original bolts, but if replacement bolts are needed due to damage/lose, then care must be

used to select the four M10 x 25mm long bolts from the hardware kit. Also contained in the hardware kit are 2 M10 x 30mm bolts for use on CT/i systems, but they could be mis-identified by Manufacturing or Service for the M10 x 25MM bolts.

If the M10 x 30MM bolts are used by mistake on a LightSpeed or HiSpeed QX/i system there is potential that they will bottom-out and compromise the mechanical joint. In this scenario there is a potential for Image Artifacts due to motion of the tube and could result in mechanical failure of the tube mounting bolts during gantry rotation.

Recently there has been several reports (3x) of this potential failure mode on LightSpeed systems. These failures are under investigation by CT Engineering and Tubes Engineering, and further information will be communicated in the near future.

2.4 Solution

- 1.) Be aware of this failure mode when troubleshooting motion Image Artifacts, and verify correct M10 x 25mm tube bolts have been installed.
- 2.) On all field tube replacements measure the attachment bolts and verify they are M10 x 25mm prior to installing.

Note: There are very clear mounting hardware instructions on a plate attached to the “tube cooling housing” and in the furnished parts list that is shipped with each Performix tube.

Appendix D - Tube Replacement: HiSpeed QX/i, LightSpeed, Discovery LS/ST

1 Introduction

1.1 Purpose

This document covers the quick replacement procedure for X-Ray Tubes. See [Figure 22](#) for an overview of this process. The sections following explain this process.

1.2 Requirements

The system must meet the following requirements to perform the Quick Tube Replacement Procedure:

- 1.) Normal Tube Change with no other system issues.
- 2.) The previous Tube was aligned correctly. (POR, BOW, CBF, ISO were in spec.)
- 3.) The Collimator and the Detector have not moved.
- 4.) There are no IQ issues.
- 5.) If kV gain pots are adjusted, you must run a full set of calibrations.

If these requirements are met, the Quick Tube Replacement Procedure can be followed. If the system does not meet these requirements, the full Tube Change Procedure described in the LightSpeed 4.X System Service Manual must be performed.

1.3 Quick Tube Replacement Process Flow - Same Tube Re-Installed

Follow this flowchart, if you remove the tube from the gantry and re-install the same tube back onto the gantry.

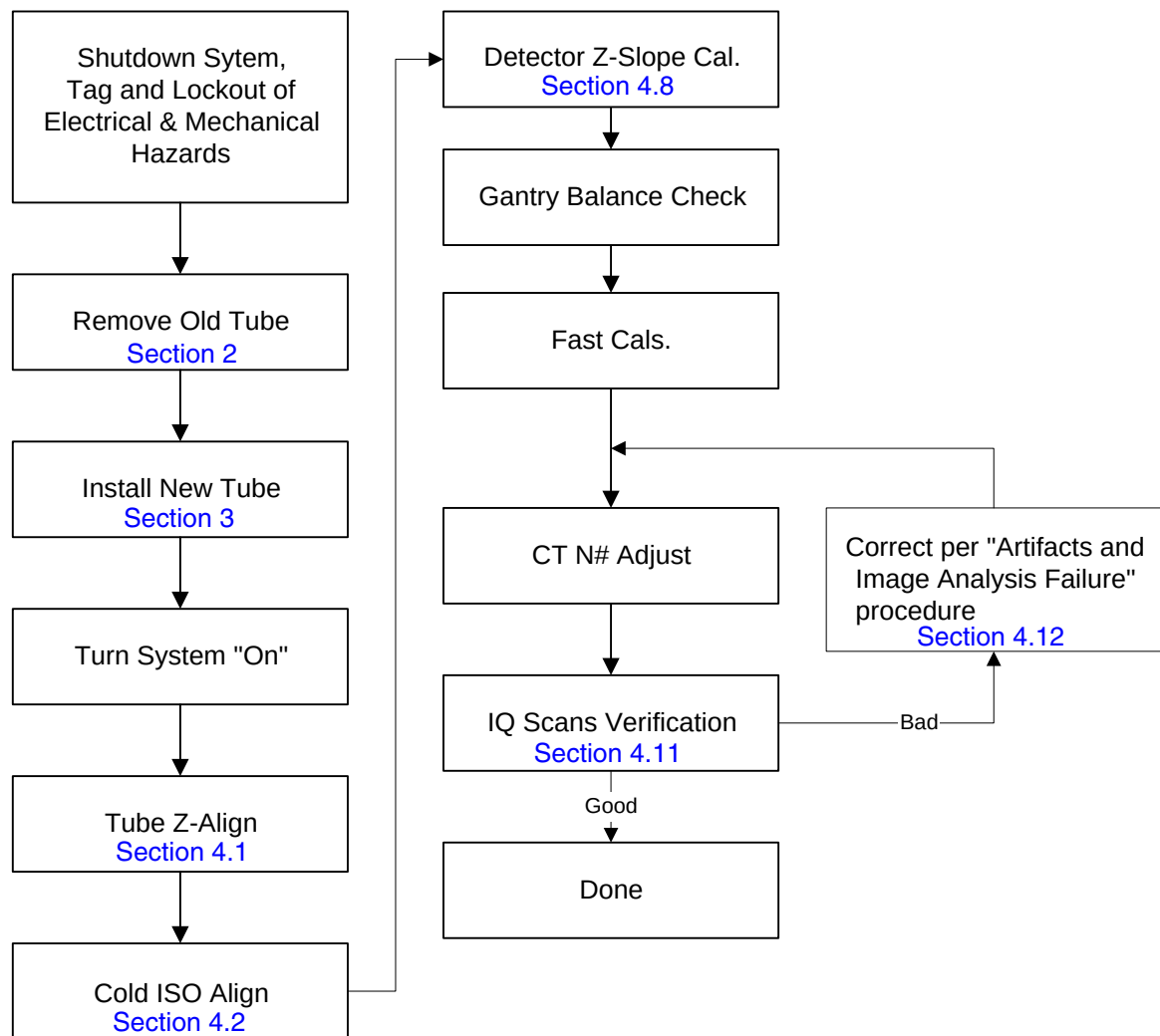


Figure 22 - Quick Tube Change Flow

1.4 Quick Tube Replacement Process Flow - Tube Replaced

Follow this procedure, if you replace the tube.

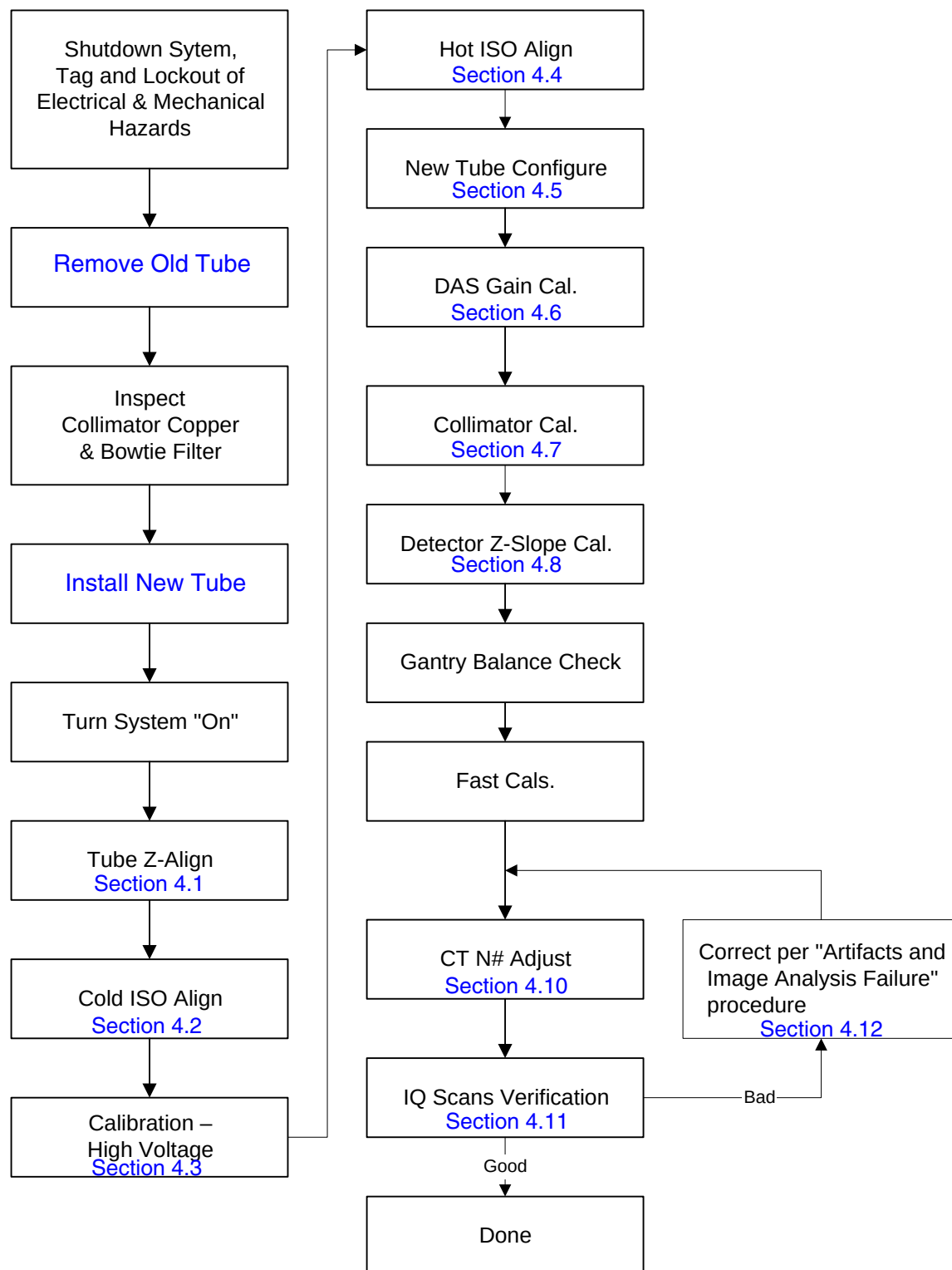


Figure 23 - Quick Tube Change Flow

2 Remove Old Tube

Before beginning this procedure, please read the safety information.

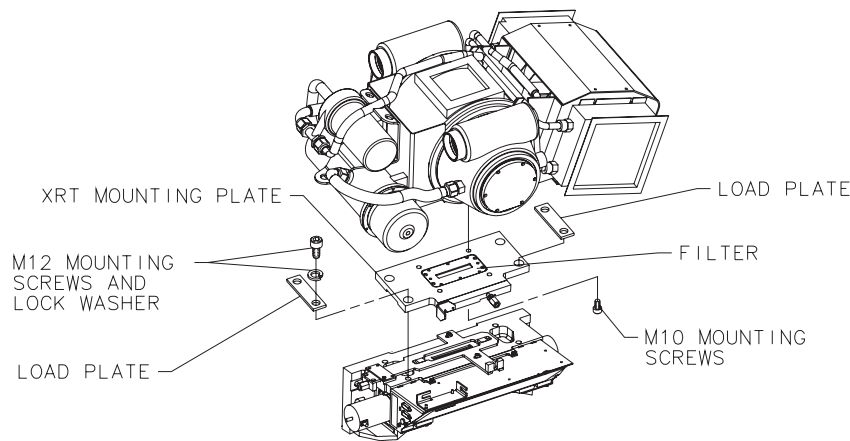


Figure 24 - Tube Removal Diagram



FOLLOW LOCKOUT/TAGOUT PROCEDURES FOR POWER & ROTATIONAL HAZARDS.

Make sure you engage the locking mechanism before you remove the tube. Failure to lock the gantry could result in injury, should the gantry suddenly move and strike you.

- 1.) Remove, and set aside, both gantry side covers, top covers, front and rear covers.
- 2.) Remove the M12 screws from the right front gantry cover mounting bracket, remove and set aside the bracket. Reference [Figure 25](#).



It might be necessary to tilt the gantry back to remove the third bolt, which is not normally installed. Remember to tilt the gantry back to zero degrees.



Figure 25 - Right Front Gantry Cover Mounting Bracket

- 3.) Turn off facility power to **PDU** and lockout/tagout.
- 4.) Turn off the **AXIAL DRIVE ENABLE** and **HVDC ENABLE** switches on the **STC** backplane.

**NOTICE**

It may be easier to loosen the M12 tube mounting bolts with the tube at about the 2 o'clock position before locking the tube at the 3 o'clock position. Simply loosen the tube mounting bolts one half (1/2) turn. Do not remove the bolts yet.

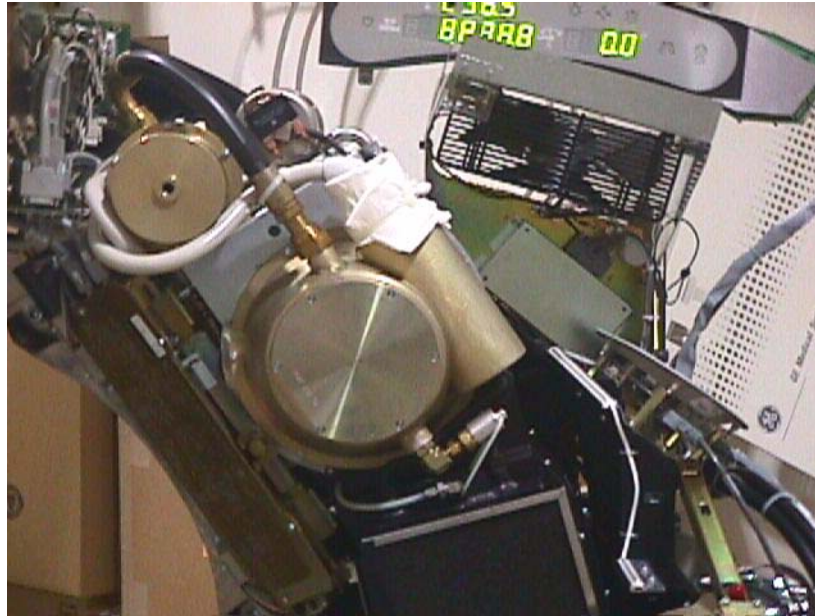


Figure 26 - Tube Angle to Loosen and Torque M12 Screws

- 5.) Rotate the Gantry until the failed tube unit reaches the 3 o'clock position.

**NOTICE**

Make sure the tube is at 90 degrees so the tube hangs at the correct engagement angle for removal and installation.

- 6.) Engage rotational lock. Check that the gantry is securely locked by attempting to rotate the gantry by hand.
- 7.) Insert the lifting post, boom and chain hoist. Reference [Figure 25](#).
- 8.) Disconnect the 12 pin tube I.D. system cable, from the top of the tube unit.
- 9.) Disconnect the 4 pin mate-n-lock pump and fan power system cable
- 10.) Disconnect the ground strap from the top of the tube unit
- 11.) Remove the anode and the cathode cable:
- Carefully cut ty-raps securing **HV** cables. Note **HV** cable routing.
 - Loosen each cable's locking ring with the spanner wrench.
 - Pull each cable terminal out of its receptacle.
 - Ground the end of the cables to the Gantry frame.
 - Wipe up any oil that drips from the cable terminal.
 - Use paper towels to soak up any oil in the wells.

**CAUTION**

Remove the mounting bars in the following (lower/upper) order to lessen the risk of injury to your hand. Keep one hand under the bolt and pressure plate while unfastening it. This is to prevent them from falling into the fan that is attached to the tube.

Note: It may be easier to tape the socket to the extension. This will prevent the socket from being dislodged on the tube radiator assembly

- 12.) The XRT is attached to the Collimator with a Tube Mount Bracket Assembly (P/N 2128696). Remove the mounting plate & XRT from the Collimator by removing the four M12 (P/N 46-328416P24) cap screws and lock washers, and two load plates (P/N 2120104), with a hex socket driver. With your hand, reach behind the radiator to the screws from either side of the XRT center section while removing the bolts with two 12 inch extensions (24 inch length) on a ratchet. **Throw these M12 bolts and washers away, as they should not be reused.**
- 13.) Carefully swing the tube clear of the gantry.

Note: Be careful not to damage the fragile copper filter or lead shield in the mounting plate for the next step.



CAUTION
Potential for IQ
artifacts

When removing the mounting plate from the tube, be careful with the Copper Filter. It should be free of debris, scratches and dust. Particles create artifacts in the image by affecting the attenuation properties of the copper filter.

- 14.) Remove the mounting plate by removing the four M10 (P/N 46-328416P20) hex head screws. **Throw these bolts and washers away, as they should not be reused.**
- 15.) **Inspect the copper filter.** The primary copper filter is easily examined, since the primary copper filter assembly is removed with the tube. The copper filter should be clean, dent and scratch free:
- If contamination is visible (see [Figure 27](#)), proceed to the cleaning process.
 - If scratches or dust are visible, the copper filter must be replaced.
 - Discoloration is acceptable.



Figure 27 - Extremely Contaminated Copper Filter

- 16.) **Inspect the Bowtie Filter.** It is possible that the Bowtie Filter is contaminated and the Primary Copper Filter is not contaminated.

Note: Perform this inspection before installing the new tube unit. Also look at the tube side of the copper filter when you are swapping the interposer plate. Remove any debris found before installing the interposer plate on the new tube unit.

The following tools are required for this inspection procedure:

- Phillips #2 Screwdriver
- Flat blade Screwdriver
- Bright Flashlight

Inspect the Bowtie Filter as follows:

- a.) Remove the Collimator Control Board Sheet Metal Cover. See [Figure 28](#).



Figure 28 - Collimator Control Board Cover

**WARNING****DO NOT ATTEMPT TO MOVE THE FILTER WITH POWER ON.**

- b.) With power removed from the gantry, use a flat blade screwdriver and position the bowtie filter assembly so it is visible through the input port or tube side of the collimator assembly.
- * CCW will move the filter into the beam. See [Figure 29](#).
 - * Do not move the filter back to the home position.

**NOTICE**
Potential for
equipment
damage

Do not force the filter if it feels stuck. Damage to the limit switch can result.
Do not move the filter more than necessary for inspection. The filter can fall off the drive screw.

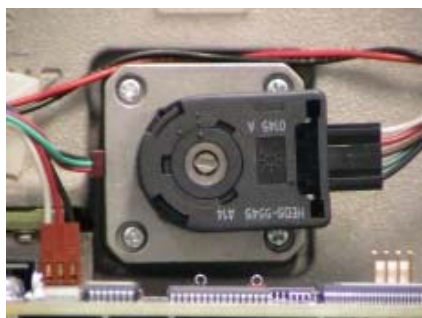


Figure 29 - Filter Position Adjuster

- c.) Using a flashlight, inspect the bowtie filter for contamination. Look through the input port or tube side of the collimator. If contamination is visible, proceed to cleaning process.

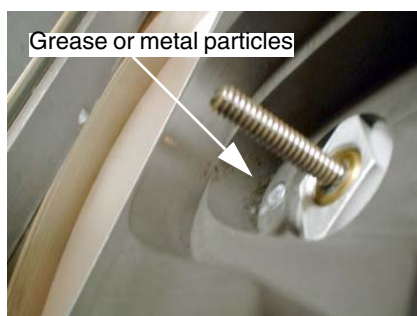


Figure 30 - Contaminated Bowtie Filter

3 Install New Tube

 **WARNING SEVERE INJURY POSSIBLE.**
TUBE CAN SEPARATE FROM GANTRY, IF NOT TORQUED CORRECTLY.
ALWAYS USE PROPER TORQUE ON ALL FASTENERS.

 **WARNING FOLLOW LOCKOUT/TAGOUT PROCEDURES FOR POWER & ROTATIONAL HAZARDS.**



- 1.) Allow the tube unit to warm to room temperature before you install it.
- 2.) Attach the mounting plate from the old tube using the **four new M10 bolts** that come with the tube. **Do NOT use Loctite.**
 - Finger tighten all four (4) bolts
 - Torque all four (4) bolts to the following specification:

15 ft-lbs	20 N-m	180 in-lbs	210 kg-cm
-----------	--------	------------	-----------

This seats each bolt, enabling you to visually ensure that the mounting holes are not stripped while applying final torque.

 **WARNING SEVERE INJURY POSSIBLE.**
IF YOU TURN THE TORQUE WRENCH MORE THAN 90° (¼ TURN) WHILE APPLYING FINAL TORQUE, THE MOUNTING THREADS ARE STRIPPED.
DO NOT USE THIS TUBE.

- Apply final torque on all four bolts:

30 ft-lbs	41 N-m	360 in-lbs	420 kg-cm
-----------	--------	------------	-----------

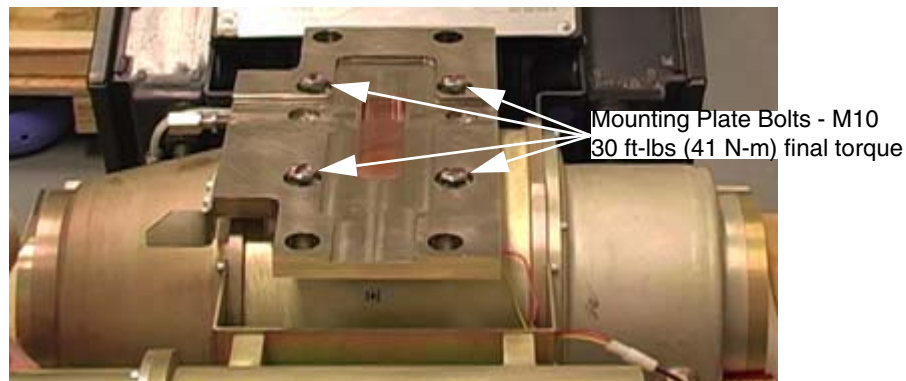


Figure 31 - Tube and Mounting Plate

 **CAUTION Potential for IQ artifacts**
When attaching the mounting plate on the tube, be careful with the Copper Filter. It should be free of debris, scratches and dust. Particles create artifacts in the image. Use a lint free wipe to clean, if needed.

- 3.) Re-check facility power and make sure it is off.

**WARNING**

SEVERE INJURY TO PATIENT COULD OCCUR.

USING THE WRONG BOLTS COULD PLACE STRESS ON BOLTS, CAUSING TUBE TO SEPARATE FROM GANTRY.

USE PROPER BOLTS, PER SYSTEM TYPE.

4.) Use the hoist to lift the new tube unit:

- Position the tube on the gantry: The “crosses” on the mounting plate and on the collimator should fit in perfectly when the tube is aligned properly.

Note:

- * To ease installation, fasten the top pressure plate to the rotating structure first. Then attach the bottom pressure plate.
- * **Use new bolts and washers from tube crate. Make sure to select the proper bolts. There are instructions in the crate, and on the tube itself.**
- Fasten the lower and upper and pressure plates to the rotating structure with the four M12(50 mm) bolts, and **torque to:**

49 ft.-lbs	66 N-m	590 in.-lbs	680 kg-cm
-------------------	---------------	--------------------	------------------

Gantry Mounting Bolts - M12
49 ft.-lbs (66 N-m) final torque

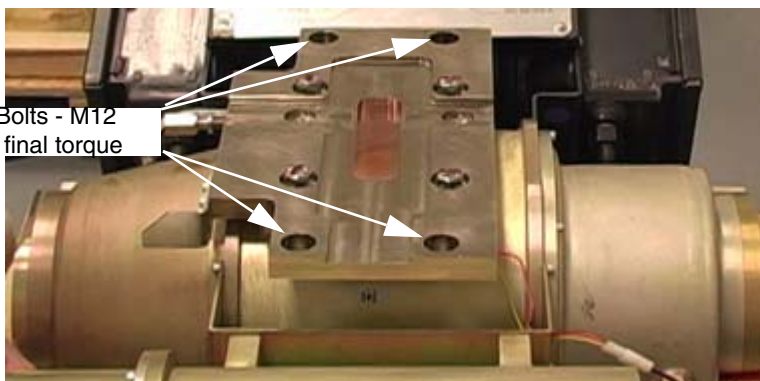


Figure 32 - Tube and Mounting Plate

- 5.) Attach the tube I.D. cable to the 12 pin mate–N–Lock connector on top of the tube.
- 6.) Attach the tube pump and fan power cable to the 4 pin mate–N–Lock connector.
- 7.) Fasten the grounding strap to the 1/4–20 ground stud on top of the tube unit.
- 8.) Remove the plastic cap plug from each cable receptacle on the tube unit.

Note:

- Take care not to lose the rubber quad rings for the High Voltage cables.
- 9.) Lightly wet the new rubber quad ring with transformer oil (917).
- 10.) Return the quad ring to its slot at the top of the receptacle retaining ring.
- 11.) Pour transformer oil (917) into the receptacle to a depth of 10 mm (0.375 in).

**NOTICE**

Incorrectly routed or secured HV cables will result in damage to the HV cables and/or other parts of the gantry.

- 12.) Be sure to route the HV cable as shown in [Figure 33](#).

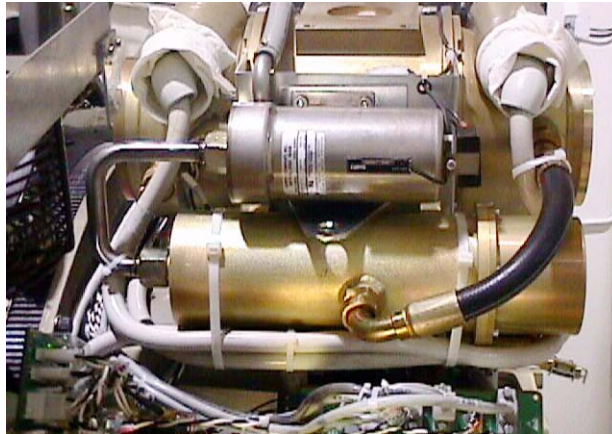


Figure 33 - HV Cables Properly Routed and Secured

- 13.) Align the cable terminal orienting key with the notch in the receptacle.
- 14.) Slowly insert the cable, to engage the connector pins, and seat the cable in the well.
- Tighten the cable locking ring.
 - Rotate the cable strain relief for a clean cable dress.

**NOTICE**

Do not over tighten the locking ring. Over tightening can deform the cable plug sealing surfaces, break the oil seal between receptacle and housing, twist the receptacle, and disrupt internal wiring.

- Use the spanner wrench to tighten the locking ring.
- Use a torque wrench to tighten the locking ring to 11.1 ft.-lbs (153 kg-cm).
- Back off on the cable locking ring without disturbing the cable plug.
- Re-tighten the locking ring, and torque to 7.1 ft-lbs (98 kg-cm).

Note: Use the spanner wrench with a torque wrench when you tighten the high-voltage cables on the tube unit.

**NOTICE**

IF YOU GET OIL ON YOUR HANDS, WASH THEM NOW.

- 15.) Carefully wipe up all excess oil.
- 16.) Secure HV Cables using Large Ty-Raps, as shown in [Figure 33](#).
- 17.) Disconnect hoist from tube and boom.
- 18.) Remove the post and boom from the gantry. Reference [Figure 25, on page 34](#).
- 19.) Check for oil leaks:
- Wrap rags or paper towels around the cable horns, and tape them into place.
 - Manually rotate the tube to the 6 o'clock position.
 - Return the tube to the 3 o'clock position
 - Remove the toweling and wipe up all excess oil.
 - Wipe off the cable horns, locking rings, and strain reliefs with a rag dampened with alcohol.
 - Repeat with a dry rag.

- Wrap the cable strain reliefs and locking rings with a single layer of absorbent paper tissue.
 - You can use two inch wide strips cut from a paper napkin.
 - Wrap the bottom edge of the paper around the top end of the cable horn, and tape it into place.
 - Extend the top edge of the paper over the top of the locking ring, and tape it to the plastic cable strain relief.
 - Remove paper after leak check.
- 20.) Install the right gantry front cover bracket. Reference [Figure 33, on page 40](#).
- 21.) Restore system power at the main disconnect panel.
- 22.) Turn on gantry **120 VAC**, **HVDC POWER** and **AXIAL DRIVE ENABLE** at the **STC** backplane. Wait at least 10 minutes to warm up the filament.

4 Alignment and Calibration

4.1 Z-Align using Beam on Window Alignment (BOW)

The purpose of the Z-Align procedure is to move the X-Ray Tube to be in alignment with the Detector and Collimator. The Detector and Collimator were aligned with the previous Tube, and Z-Align will correctly place the new Tube in the Z-Direction.

Note: For the BOW tool to work correctly, the X-Ray Tube needs to be close to proper alignment with the Collimator and Detector. If the Tube is not close to where it needs to be, the BOW tool will give erroneous data back. A POR scan and corresponding Tube movement can correct this problem.

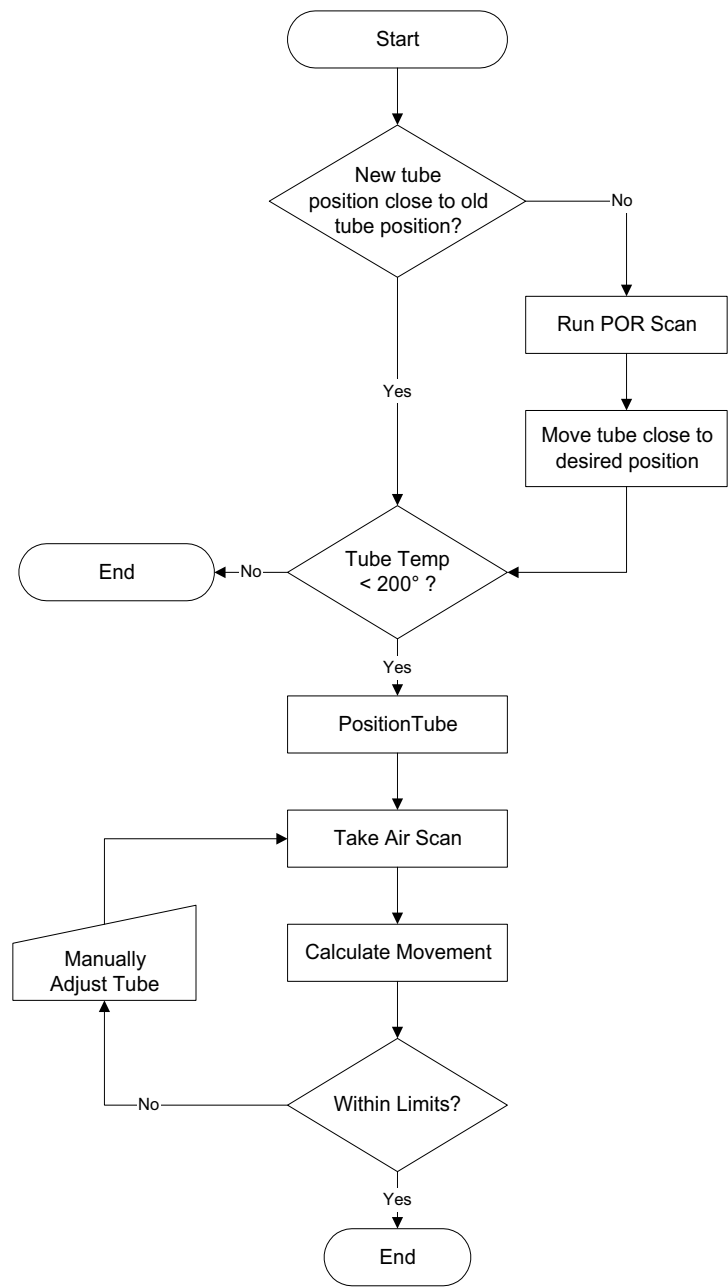


Figure 34 - BOW Alignment Process Flow

4.1.1 Verify Tube Temperature < 200° C

- 1.) Click DAILY PREP.
- 2.) Click TUBE WARM-UP.
- 3.) Click ACCEPT in pop-up window.
- 4.) Click PAUSE.
- 5.) Click CANCEL (do not QUIT Daily Prep).



NOTICE

Do not start X-ray exposures. Starting tube warm-up will result in additional tube cooling wait times.

- 6.) Open system error log (gesys_suite.log).
- 7.) Click LAST PAGE.

Look for this type of entry:

This entry is created when Pause is clicked.

```
bay57 dailyPrepRx
```

```
StateMachineEventNotify.c 418
```

```
Tube temperature before Cold Tube Warmup is 315.15 degrees Celsius.
```

This entry is created when Cancel is clicked.

```
bay57 dailyPrepRx
```

```
StateMachineEventNotify.c 411
```

```
Tube temperature after Cold Tube Warmup is 312.89 degrees Celsius.
```

```
Start Tube Alignments when 200.00 degrees Celsius is reported.
```

```
bay57 dailyPrepRx
```

```
StateMachineEventNotify.c 418
```

```
Tube temperature before Cold Tube Warmup is 200.00 degrees Celsius.
```

- 8.) Repeat steps 3 through 5 as necessary to generate updates in the gesyslog.
- 9.) When tube temperature shows 200.00 Celsius, click QUIT to exit Daily Prep.

4.1.2 Accessing the Software

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT
- 3.) Select TUBE CHANGE
- 4.) Select BOW ALIGNMENT

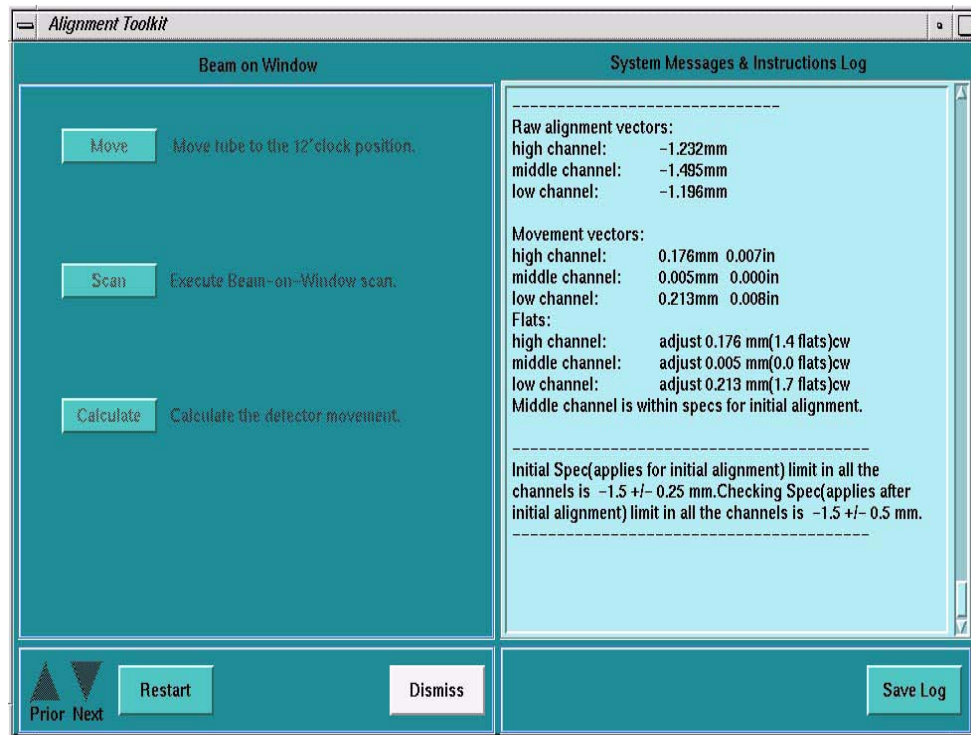


Figure 35 - Screen Shot for Beam on Window Alignment

4.1.3 Adjustment Procedure

- 1.) Place a dial gauge in the POR Dial Mount location.
- 2.) Select MOVE to send the tube to the 12 o'clock position.
- 3.) Acquire a Beam on Window Scan.
- 4.) Select CALCULATE.
- 5.) If adjustments are required, calculate the Tube Z-Movement to bring BOW into spec.
BOW Movement vectors:
 - BOW high channel: -0.446mm -0.018in
 - **middle channel: -0.653mm -0.026in**
 - low channel: -0.611mm -0.024in
- 6.) After Selecting the CALCULATE button from the BOW Window, the GUI will display the BOW Raw Alignment Vectors, BOW Movement Vectors and the Flats Movement Vector.
- 7.) If the middle channel flats adjustment is more than 10 (either direction), run one POR adjustment to get the tube close. Then, proceed with Z-align, from the beginning. POR adjustment directions can be found in the applicable *System Service Manual*.
- 8.) Tube Z-Movement Value = BOW Middle Channel Movement Vector divided by five.
- 9.) Tube Z-Movement Direction:
 - a.) If the BOW Movement Channel value is Positive (+) move the Tube towards the REAR of

the Gantry. Turn the POR nut CCW.

- b.) If the BOW Movement Channel value is Negative (-) move the Tube towards the TABLE. Turn the POR nut CW.

Example: From the sample data above, the BOW Movement Vector for the Middle Channel is **-0.653mm or -0.026in**.

- Tube Z-Movement Value (mm) = $-0.653 \text{ mm} / 5 = -0.131 \text{ mm}$ OR
- Tube Z-Movement Value (in) = $-0.026 \text{ in} / 5 = -0.0052 \text{ in}$.
- Tube Z-Movement Dir. negative = Towards the Table, turn the POR nut CW.

- 10.) Loosen the Tube Mounting Bolts.
- 11.) Move the Tube the desired Direction and Distance. Be as accurate as possible to avoid rescans.
- 12.) Tighten the Tube Mounting Bolts (49 ft lb)
- 13.) Select MOVE to send the tube to the 12 o'clock position.
- 14.) Acquire a Beam on Window Scan.
- 15.) Select CALCULATE
- 16.) If the Values are within spec, move on. If not, return to step 7.

Note: The specs for **BOW** are checked by the software. If an adjustment is needed for the first **BOW** scan, make adjustments per procedure. In the verification scan, after ISO alignments, the spec is different than the software version because the tube is warm. The spec is $-1.5 \pm 0.5\text{mm}$.

4.2 Cold ISO Alignment

Cold ISO Alignment is done before the tube is heated by Generator calibrations. Hot ISO is done later. For details on Hot ISO, see [Figure 4.4, on page 57](#).

Note: Use two dial gauges for ISO adjustment, to ensure POR accuracy while the tube is loose. This saves valuable time when calibrating for good image quality.

4.2.1 Verify Tube Temperature < 200° C

For details on how to verify Tube Temperature, see the procedure in section 4.1.1.

4.2.2 Accessing the Software

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT
- 3.) Select TUBE CHANGE

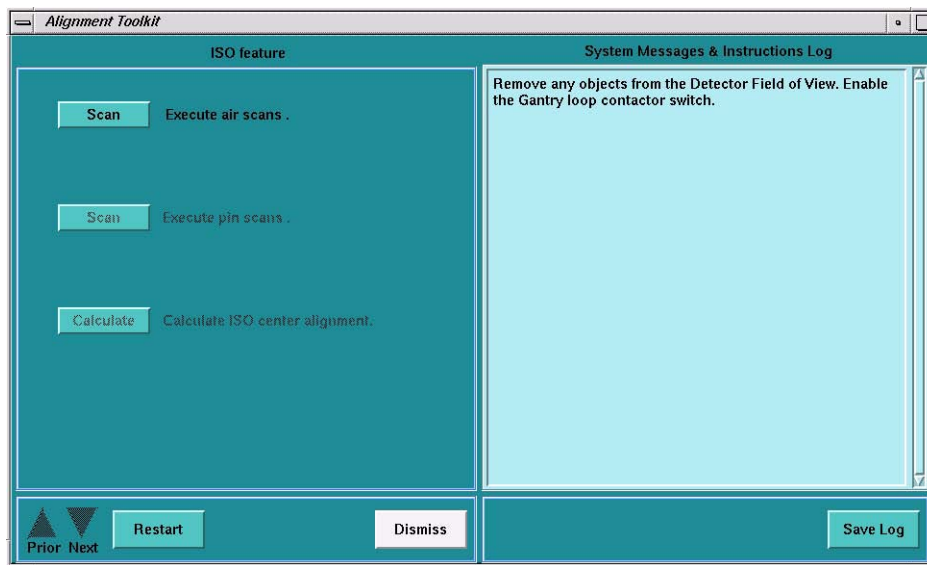
4.) Select ISO ALIGNMENT

Figure 36 - ISO Alignment Menu

4.2.3 ISO Adjustment Procedure

- 1.) Execute Air scan (small spot).
- 2.) Execute Air scan (large spot).
- 3.) Place the 1/8 inch screw driver on the phantom holder (should be pointing into the Z direction).
- 4.) Execute pin scan (small spot).
- 5.) Execute pin scan (large spot).
- 6.) **CALCULATE** ISO center alignment.
- 7.) Mount two dial indicators on the tube assembly. (Figure 37.) Make sure that you zero them.



Figure 37 - ISO Dial Gauge Mounting Location

- 8.) Loosen the 4 M-12 bolts on the tube assembly a half turn.

- 9.) Adjust the tube UP / DOWN as indicated by the calculation. The adjustment bolt is located on the top of the tube - Please see [Figure 38](#).

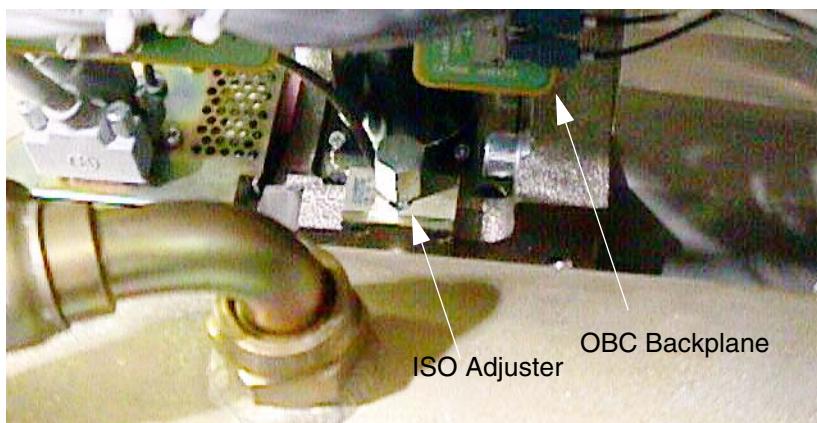


Figure 38 - ISO Alignment Adjuster

- 10.) Tighten the four (4) M12 bolts and verify ISO dial gauge still reads the correct adjustment value, and POR Gauge indicates no POR movement.
- 11.) Repeat steps 1 through 6.
- 12.) If the adjustments are within limit proceed to the next step, otherwise go to step 9.
- 13.) Tighten the four (4) M12 bolts. Torque to 49 ft-lbs (66.4 Nm).
- 14.) If ISO adjustments were made, recheck BOW to verify that the Beam-on-Window values are within the drift spec of $-1.5\text{mm} \pm 0.5\text{mm}$.
- If BOW is within spec, DO NOT adjust anything
 - If BOW is out of spec, Z-align, ISO and BOW recheck will need to be done.

4.3 Calibration - High Voltage

4.3.1 Access HV Maintenance through Service Desktop

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select GENERATOR CHARACTERIZATION.

4.3.2 Generator Characterization

Use the Generator Characterization Program to update the “small spot” and “large spot” characterization files, to provide a starting point for the closed loop mode of the generator. This iterative process requires several scans at a different KV/MA/spot size. It calculates corrections, repeats the scan until the results fall within tolerance, then updates the characterization file.

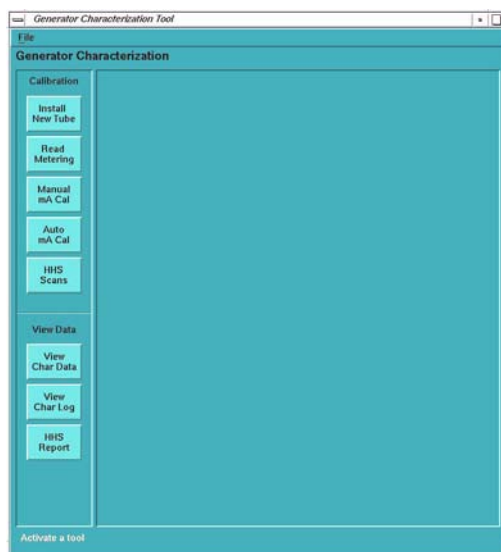


Figure 39 - Proprietary Generator Characterization Menu Screen

Real Time Information
Patient Handling
Scanning
Delay Timer
Auto mA Calibration
80 kV, 100 mA, 0.1 sec, 0.0mm
0 of 8 Tube calibration stations have completed
Cancel Pause Resume

Figure 40 - Auto mA Calibration Status Screen

4.3.3 Verify kV Meter

This section describes the calibration check of system internal kV metering circuits.

- 1.) Select READ METERING.
- 2.) Select RUN to start the test. During the test, the firmware reads the metering circuits in the OFF state, then reads the metering circuits in the ON state, and finally reports the readings to the display.
- 3.) Compare the data in the "Delta" column on the Read Meter screen (Figure 9-4) to the data in the "Limit" column.

Note: "Delta" = DVM - A/D

CIRCUIT "OFF"	CIRCUIT "ON"
Anode kV = 0 ± 0.5	Anode kV = 50 ± 7.5
Cathode kV = 0 ± 0.5	Cathode kV = 50 ± 7.5
Total kV = 0 ± 0.5	Total kV = 100 ± 15.0

Table 4 - Generator Characterization Test Specifications



Figure 41 - Proprietary Read Metering Screen

4.3.4 Verify mA Meter – HHS Requirement

WARNING

WHEN SERVICING THE GANTRY:

- **NEVER PUT ANY BODY PART INTO THE GANTRY WITHOUT FIRST DISABLING THE AXIAL DRIVE AND RE-VERIFYING (CHECK TWICE) THAT IT IS DISABLED.**
- **ENSURE THAT THE DRIVE STATUS LED (Figure 42) IS NOT LIT. DO NOT SERVICE THE GANTRY IF THIS LED IS ON.**

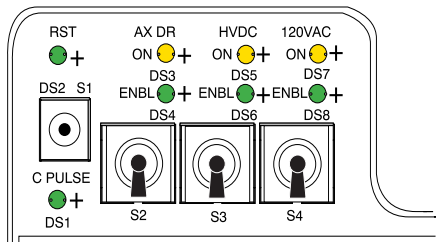


Figure 42 - STC Backplane Service Switches

This section describes the calibration check system internal mA metering circuits.

- 1.) Inside the Gantry:
 - a.) Switch OFF the HVDC ENABLE on **STC** backplane.
 - b.) Switch OFF the AXIAL DRIVE ENABLE on **STC** backplane.
 - c.) Rotate the Anode tank to the 2 o'clock position.
 - d.) Engage the gantry rotational lock.
- 2.) Select SERVICE DESKTOP.
- 3.) Select REPLACEMENT.
- 4.) Select HV SUBSYSTEM CHANGE.
- 5.) Select GENERATOR CHARACTERIZATION.
- 6.) Select READ METERING.

Note: On the display, enter a time delay in seconds, to provide enough time to walk from the console to the DVM, and record the reading. The test will not begin until this time delay expires. Once it begins, the test enables the meter circuit for only 4 seconds.

- 7.) Use a DVM as an mA meter; connect it to the hardware on the anode side:
 - a.) Connect the black lead to TP8 (ACAL1) on the mA board.
 - b.) Connect the red lead to TP11 (ACAL2) on the mA board.

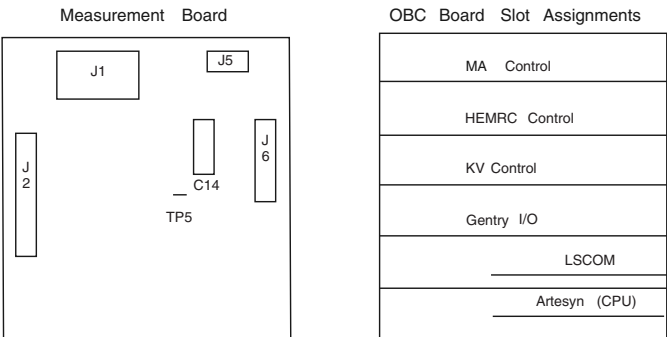


Figure 43 - Tank Measurement Board

- 8.) On the Display, select the ACCEPT button.
- 9.) Record the displayed, and measured, Anode mA values for "Circuit OFF" and "Circuit ON".
- Note: Your system has the test wire to TP5 included in the harness, the Cathode side should read approximately 19 mA during "Circuit On".
- 10.) Disconnect the test equipment from the Anode side, if used.
- Note: When you exit Generator Characterization, this test may generate
kV board tube spit counter = x error messages.
- 11.) Use a DVM as an mA meter:
 - a.) Connect the black lead to TP9 (CCAL1) on the mA board.
 - b.) Connect the red lead to TP14 (CCAL2) on the mA board.
- 12.) On the Display, Select the ACCEPT button.
- 13.) Record the displayed, and measured, Cathode mA values for "Circuit OFF" and "Circuit ON".
- Note: Your system has the test wire to TP5 included in the harness, the Anode side should read approximately 20mA during "Circuit On".
- 14.) Disconnect the test equipment from the Cathode side if used.
- Note: When you exit Generator Characterization, this test may generate
kV board tube spit counter = x error messages.

4.3.5 KV Gain Pots Adjustment – HHS Requirement

Install HV Divider

- 1.) Inside the Gantry on the **STC** backplane:
 - a.) Switch OFF the HVDC ENABLE.
 - b.) Switch OFF the AXIAL DRIVE ENABLE.
 - c.) Rotate the Tube to the 3 o'clock position
 - d.) Engage the gantry rotational lock.
 - e.) Switch OFF the 120 VAC GANTRY POWER.
 - f.) Install the **HV** Divider between Tube and Tanks.
- Note: Place the **HV** Divider on a table or tube hoist, so the cables reach the tube.
- 2.) Add a ground wire (minimum size of AWG 12) from Tube ground to bleeder ground. Refer to [Figure 44](#).



CAUTION

Potential Electrical Hazard. Performix tube unit MUST be grounded to the gantry during testing.

- 3.) Switch ON the 120 VAC GANTRY POWER.
- 4.) Switch ON the HVDC ENABLE.
- 5.) Press the  button, on the gantry control panel, turning ON the drives power.
- Note: If the gantry covers are removed press the RESET BUTTON on the **STC** backplane to turn ON Drives power.
- 6.) Reset the hardware.



NOTICE
Potential for
tube damage

Incorrect installation of anode and cathode HV cables can destroy the Performix tube unit.

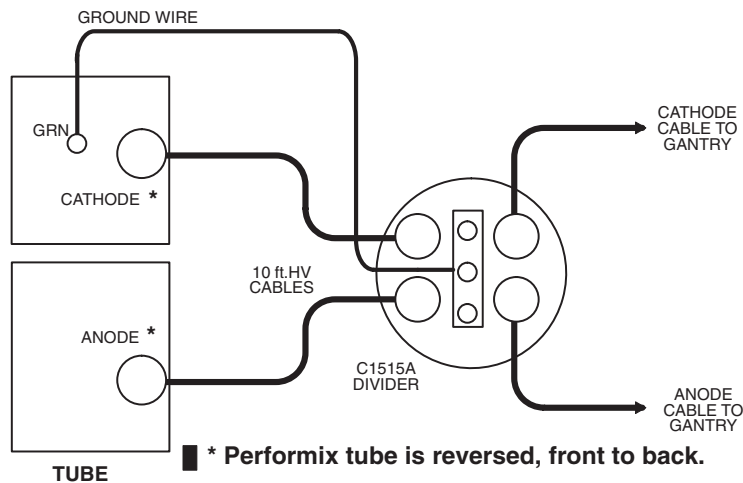


Figure 44 - HV Divider Installation

Set Up Instrumentation

Use an oscilloscope with 10X probes

- 1.) Use the Gantry Service Outlet to provide 120 Vac power for the scope. This will reduce noise on the scope waveform.
- 2.) Connect **channel one** to the **anode** output of the divider. Connect the scope ground to bleeder ground.
- 3.) Connect **channel two** to the **cathode** output of the divider. Connect the scope ground to bleeder ground.

Note: In order to minimize bleeder-induced ripple on the kV waveform, connect a 30 foot Belden shielded twisted pair cable between the scope probes and the bleeder.

- 4.) Trigger channel one, positive, DC couple, trigger mode normal.
- 5.) Channel one and two, 10v/div, time base 200ms.
- 6.) Invert Channel two.

Calibrate the Cathode

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select BLEEDER SETUP (DDC)
- 5.) Verify/Set-up the following **DDC** parameters:

- <u>STATIC X-RAY ON</u>	Interscan Delay <u>2.00</u>
- <u>1 SECOND</u>	DAS Gain <u>31</u>
- <u>1 SCAN</u>	Gantry Tilt <u>0.0</u>
- <u>FOCAL SPOT LARGE</u>	Trigger Rate <u>984</u>
- <u>100 KV</u>	Filter <u>BLOCKED</u>
- <u>50 MA</u>	Calibration Vector <u>NONE</u>
- <u>MONITOR ENABLE</u>	

- 6.) Select ACCEPT RX. The Computer Displayed reading specification for the Cathode kV and Anode kV equals 50 ± 0.5 kV.
- Note: If you use scope cursors to window the trace, position the Left Vertical Cursor to the Right of the Rising Edge of the waveform. Position the Right Vertical Cursor to the Left of the Falling Edge of the Waveform.
- 7.) Adjust the Cathode pot on the kV board, until the scope reading for the Cathode kV, and the displayed reading for the Cathode kV in the message log, fall within ± 0.5 kV of each other.
- 8.) Use the pot, labeled CAKV, R316, on the kV board, to adjust the scope reading.
- CCW decreases the scope kV.
 - CW increases the scope kV.
 - 1/2 turn equals approximately 0.5 kV.
- 9.) Record the results on **FORM 4879**.

Calibrate the Anode

- 1.) Select SERVICE DESKTOP.
 - 2.) Select REPLACEMENT.
 - 3.) Select HV SUBSYSTEM CHANGE.
 - 4.) Select BLEEDER SETUP (DDC).
 - 5.) Verify/Set-up the following **DDC** parameters:

- <u>STATIC X-RAY ON</u>	Interscan Delay <u>2.00</u>
- <u>1 SECOND</u>	DAS Gain <u>31</u>
- <u>1 SCAN</u>	Gantry Tilt <u>0.0</u>
- <u>FOCAL SPOT LARGE</u>	Trigger Rate <u>984</u>
- <u>100 KV</u>	Filter <u>BLOCKED</u>
- <u>50 MA</u>	Calibration Vector <u>NONE</u>
- <u>MONITOR ENABLE</u>	
 - 6.) Select ACCEPT RX. The Computer Displayed reading specification for the Cathode kV and Anode kV is 50 ± 0.5 kV.
- Note: If you use scope cursors to window the trace, position the Left Vertical Cursor to the Right of the Rising Edge of the waveform. Position the Right Vertical Cursor to the Left of the Falling Edge of the Waveform.
- 7.) Adjust the Anode pot on the kV board, until the scope reading for the Anode kV, and the displayed reading for the Anode kV in the message log, fall within ± 0.5 kV of each other.
 - 8.) Use the pot, labeled ANKV, R318, on the kV board, to adjust the scope reading.
 - CCW decreases the scope kV.
 - CW increases the scope kV.
 - 1/2 turn is approximately 0.5 kV. - 9.) Record the results on **FORM 4879**.

Measure Total kV

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select BLEEDER SETUP (DDC).

- 5.) Verify/Set-up the following **DDC** parameters:

- <u>STATIC X-RAY ON</u>	Interscan Delay <u>2.00</u>
- <u>1 SECOND</u>	DAS Gain <u>31</u>
- <u>1 SCAN</u>	Gantry Tilt <u>0.0</u>
- <u>FOCAL SPOT LARGE</u>	Trigger Rate <u>984</u>
- <u>100 KV</u>	Filter <u>BLOCKED</u>
- <u>50 MA</u>	Calibration Vector <u>NONE</u>
- <u>MONITOR ENABLE</u>	
- 6.) Change the oscilloscope to add ch.1 and ch.2, to read total kV from the **HV** divider.
- 7.) Channel one and two, 20v/div, time base 200ms, trigger channel. one, positive.
- 8.) Select ACCEPT RX.
- 9.) Record the scope reading, and the Average. kV displayed in the message log, in **FORM 4879**.
- 10.) Display the Generator Characterization menu.
- 11.) Toggle the soft-key MONITOR ENABLE **OFF**, so the message log no longer displays kV and mA readings.

Verify kV Meter

Use this procedure to verify the calibration of the internal kV metering circuits.

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select GENERATOR CHARACTERIZATION.
- 5.) Select READ METERING.
- 6.) Select ACCEPT.
 - The test begins after the time delay expires.
 - Once the test begins, the software enables the meter circuit for 4 seconds.
- 7.) Record the displayed Anode kV, Cathode kV and Total kV values in the **FORM 4879** "Circuit OFF" and "Circuit ON" table.
- 8.) Select BACKUP.

Remove the External HV Divider

- Note:  Press the button, on the gantry control panel, turning **OFF** the drives power.
If the gantry covers are removed, proceed to step 2.
- 2.) Inside the Gantry on the **STC** backplane:
 - a.) Switch OFF the HVDC ENABLE.
 - b.) Switch OFF the AXIAL DRIVE ENABLE.
 - c.) Switch OFF the 120 VAC GANTRY POWER.
 - d.) Remove the **HV** Divider between the Tube and Tanks ([Figure 44, on page 52](#)).
 - e.) Reconnect the **HV** cables for normal operation.



NOTICE

Incorrect installation of anode and cathode HV cables can destroy the Performix tube.

- f.) Re-apply paper toweling around tube locking ring to absorb excess oil.

- g.) Disengage the gantry rotational lock.
- a.) Switch ON the HVDC ENABLE.
- b.) Switch ON the AXIAL DRIVE ENABLE.
- c.) Switch ON the 120 VAC GANTRY POWER.

- 3.) Press the  button, on the gantry control panel, turning ON the drives power.

Note: If the gantry covers are removed, press the RESET BUTTON on the **STC** backplane to turn ON Drives power.

- 4.) Reset the hardware.

Install New Tube Program – HHS Requirement

Use this program to complete Auto mA Cal on a new tube. *Run this program only on a new tube.* Refer to Figure 9-5.

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select GENERATOR CHARACTERIZATION.
- 5.) Select the INSTALL NEW TUBE.

Note: The system automatically warms up the tube.

- 6.) The system prompts you with the tube type. Verify the number corresponds to your tube type; answer **Y** or **N**.

SOFTWARE TOKEN	HOUSING NUMBER	INSERT NUMBER
12-MX_135CT	46-274800G1	46-274600G1
13-MX_165CT	46-309500G2	46-309300G1
14-MX_165CT_I	46-309500G2	46-309300G2
15-MX_200CT	2137130-2	2120785

Table 5 - Tube Type Table (SW tokens for various Housings and Inserts)

- 7.) Press START SCAN when it flashes, to automatically run the program and update the display:
seed filament current shift scans

Auto mA Calibration

Run this program when you replace the X-Ray tube, or the system requires re-calibration.

Note: If you have just completed running the Install New Tube Program in the previous section, you do NOT need to run Auto mA Cal.

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select GENERATOR CHARACTERIZATION.
- 5.) Select AUTO MA CAL.

Note: The software automatically warms up the tube.

- 6.) Press START SCAN when it flashes, to automatically run the program and update the display:
 - Ductility warm-up -
 - Auto mA Cal -
- 7.) The system displays the final filament currents on the screen.

kV AND mA Verification Data Sheet – HHS Requirement

Access the software and perform the following:

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select TUBE CHANGE.
- 4.) Select HHS SCANS.

When Scans are complete:

- 5.) Select HHS REPORT.

Insure all kV and mA stations meet specifications found in form 4879. (Form can be found in the appropriate system Advanced Service manual.)

Note: If your system has InSite connectivity, you can have a copy of the HHS report e-mailed to your laptop for printing by sending a sweep report to the autoSC.

Send an e-mail in the following format:

TO: autosc (MED) OR autosc@olc.med.ge.com

SUBJECT: sweep

Message content:

type = get_hhs_report

system id = xxxxx

4.4 HOT ISO Alignment

4.4.1 Accessing the Software

- 1.) Select SERVICE DESKTOP.
- 2.) Select CALIBRATION.
- 3.) Select HOT ISO ALIGNMENT.

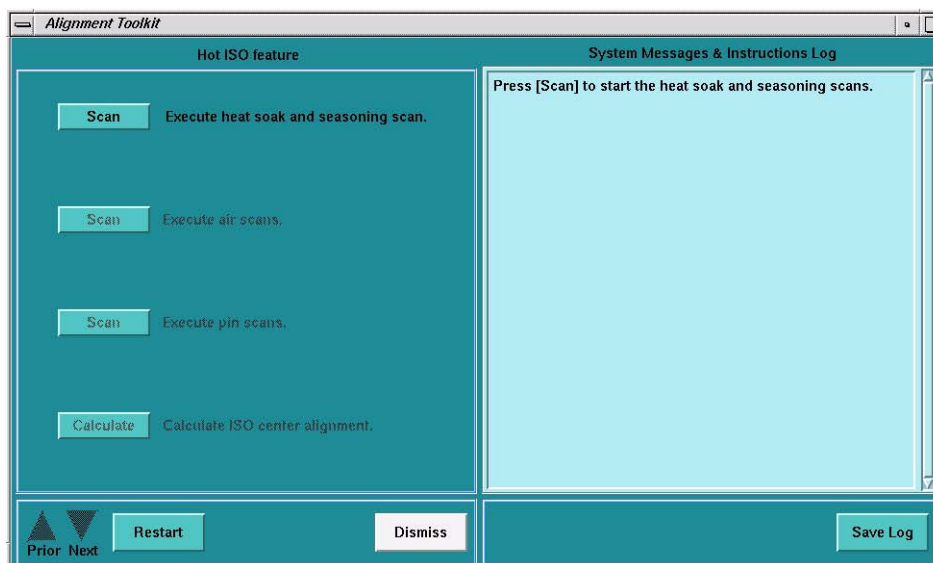


Figure 45 - HOT ISO Screen

4.4.2 Adjustment Procedure

- 1.) Execute the Heat Soak and Seasoning scan.
- 2.) Execute Air Scan (small spot).
- 3.) Execute Air Scan (large spot).
- 4.) Place the 1/8 inch screw driver on the phantom holder (should be pointing into the Z direction).
- 5.) Execute Pin Scan (small spot).
- 6.) Execute Pin Scan (Large spot)
- 7.) Calculate correction factor with Calculate button.

Note: No movement of the tube is required. This is a software correction done automatically.

4.5 New Tube Configure

Access the software and perform the following:

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select TUBE CHANGE.
- 4.) Select NEW TUBE CONFIG.
- 5.) Select SCAN HARDWARE RESET and reset application SCAN.

4.6 DAS Gain Calibration

- 1.) Perform **DAS** Gain Calibration by selecting the function from Scanner Utilities (**left head**).
- 2.) Select SCANNER UTILITIES, and Select DAS GAIN CALIBRATION.
- 3.) Ensure that there is nothing in the x-ray beam and continue.
- 4.) The system will now perform a Mylar window check and provide the appropriate messages if the window should need cleaning.
- 5.) Upon completion of the Mylar window scans the system will now take 31 scans and save the results in the Calibration Data Base. The system will provide the appropriate messages if the calibration should fail.

4.7 Collimator Calibration

- 1.) Perform Collimator Calibration by selecting the function from Scanner Utilities (**left head**).
- 2.) Select SCANNER UTILITIES, and Select COLLIMATOR CALIBRATION.
- 3.) Ensure that there is nothing in the x-ray beam and continue.
- 4.) The system will now perform a Mylar window check if needed and provide the appropriate messages if the window should need cleaning.
- 5.) Upon completion of the Mylar window scans the system will now take 8 scans and save the results in the Calibration Data Base. The system will provide the appropriate messages if the calibration should fail.

4.8 Detector Z-Slope Calibration

- 1.) Perform Z-Slope Calibration by selecting the function from Scanner Utilities.
- 2.) Select SCANNER UTILITIES, and select Z-SLOPE CALIBRATION.
- 3.) Ensure there is nothing in the x-ray beam and continue.
- 4.) The system will now scan and provide the appropriate messages if the calibration should fail.

4.9 Calibration Process



CAUTION
Potential for
IQ Artifacts

If kV Gain Pots were adjusted in section 4.3.5, you cannot use Fastcal. A full set of calibrations must be performed.

- 1.) Perform a Gantry Balance check.
 - a.) SERVICE DESKTOP > REPLACEMENT > GANTRY SERVICE BALANCE
 - b.) Check that the tube counterbalance weights are installed and bolts are secure.
 - c.) Complete the Gantry balance check.
- 2.) If this test does not pass, it is not recommended that you perform calibrations until the balance issue is resolved. Refer to the balance procedure in the Gantry Section of the appropriate Advanced Service Manual.
- 3.) Bring up the main menu: It is represented as an icon located on the bottom of the screen labeled as "Daily Prep". Click the on-screen DAILY PREP button (**left head**).



CAUTION
Potential for
IQ Artifacts

If Fastcal has been run in the past 48 hours, only one filament will be selected. In this case, Fastcal will need to be run twice in order to calibrate both filament sections. It is suggested that you watch the Fastcal scan list as scans are being taken to insure that both filaments are being scanned.

Note: To check when the last Fastcal was run:

- a.) Open a unix shell.
- b.) `cd /usr/g/service/state`
- c.) `tail -50 ssw.calreport.hist`
- 4.) Select and run FASTCALIBRATION.
- 5.) The system will run through the Normal Fast Calibration Procedure.

4.10 CT N# Adjust

- 1.) After the Fast Calibration is run, run CT N# Adjust Scans.
- 2.) Select SCANNER UTILITIES, and select AUTO CT NUMBER ADJUST.
- 3.) Place the 20 cm water phantom on the phantom holder and center the phantom.
- 4.) Run the CT N# Adjust scans.



CAUTION
Potential for IQ
artifacts

Because **DETAILED CALIBRATIONS** were not run, it is **IMPERATIVE** to visually inspect all **CT N# Adjust Images** for artifacts.

- 5.) After all Scans have completed, visually inspect all images for artifacts. If artifacts are seen, take note of the scan configuration, i.e. 4 x 2.5 / 200 mA / 120 kV / 2s

4.11 Quick Tube Change IQ Scans and Image Analysis

4.11.1 Quick Tube Change IQ Scans – Large FOV

Purpose: Because Auto CT Number Adjust scans only look at the small field of view, there needs to be a series of scans taken with the large field of view. The 35cm and 48cm Phantoms will be used. Here is a listing of scans, and the procedure to run Image Analysis.

TEST	PROTOCOL	SERIES DESCRIPTION	4-SLICE	8-SLICE	16-SLICE
QA3 - 120kV - 10mm - 2i	Image Series QA	QA#3/Small/1.0s	X		
QA3 - 120kV - 8x2.50		QA#3/Small/8x2.5		X	
QA3 - 120kV - 16x0.625		QA#3/Small/16x0.625			X
QA3 - thintwin or 0.625/2i		QA#3/Small/2x0.625/2s	X	X	X
P35 - 4x1.25	Image Series 35cm Poly	35cm/4x1.25/2s	X		
P35 - 8x1.25		35cm/8x1.25/2s		X	
P35 - 16x1.25		35cm/16x1.25/2s			X
P48 - 4x2.5	Image Series 48cm Poly	48cm/4x2.5mm/1s	X		
P48 - 8x2.5/120/200/1s		48cm/8x2.5mm/1s		X	X

Table 6 - Quick Tube Change IQ Scans

- 1.) The phantom under test must be centered before starting scanning.
- 2.) Select SCANNER UTILITIES and CENTER PHANTOM.
- 3.) Repeat using the Center Phantom function until the phantom is centered.

- 4.) Exit the Center Phantom Utility and The Scanner Utilities Screen.
- 5.) Select NEW PATIENT.
- 6.) Enter a PATIENT ID, and PATIENT NAME.
- 7.) Select the SERVICE tab from the Anatomical Selector.
- 8.) Select MANUFACTURING.
- 9.) Select the Protocol being investigated (Image Series QA, Image Series 35 cm Poly or Image Series 48 cm).
- 10.) Hit the NEXT SERIES button until the desired system configuration appears.
- 11.) Scan the series and move on to the next desired system configuration.
- 12.) When the IQ scans are done for a phantom, select END EXAM.
- 13.) Return to Step 1 and repeat for all phantoms being scanned.
- 14.) When all Phantoms have been scanned from the Service Desktop, select IMAGE QUALITY and IMAGE ANALYSIS.
- 15.) Select the small AUTO 1X button at the top of the screen.
- 16.) Select the pull down menu AUTO 1X, select the Phantom to Analyze (ImgSer 48cm or ImgSer 35cm), and select the Scan Configuration to Analyze, and select the AUTO button.
- 17.) In the SORT WINDOW, make sure the exam being Analyzed is highlighted.
- 18.) Back in the Image Analysis Window click on the ACCEPT button.
- 19.) A PASS/FAIL window will appear, along with details of what Passed and Failed.
- 20.) If the scan passed, that system configuration is in an acceptable state. Repeat steps 16 through 19 until all scans have been analyzed.
- 21.) If any scans fail, take note of the system configuration.

4.11.2 Image Quality Check for Form 4879 - HHS Requirement

In addition to the Quick Tube Change IQ Scans, you will also need to refer to the FORM 4879 that is appropriate for your scanner and scan the phantoms required at the prescribed techniques in order to complete FORM 4879.

4.12 For Artifacts and Image Analysis Failures

Purpose: If there were no Artifacts or Image Analysis Failures, the Quick Tube Replacement Procedure is complete. Follow the following instructions if you have Artifacts or Failures.

- 1.) Compile your notes from CT N Adjust Artifacts and Image Analysis Failures.
- 2.) Select SCANNER UTILITIES and DETAILED CALIBRATION.
- 3.) A Gantry Balance Check will run.
- 4.) From the Detailed Calibration list, select only the scan configurations that failed.

Example: There was a CT Number Adjust Artifact at 120kV with a 4x1.25 scan configuration, and there was an Image Analysis Failure at 120kV at 4x2.5, the 120KV button and the 4x1.25 and 4x2.5 slice collimation buttons would be selected. Leave all of the SFOV and FOCAL SPOT buttons selected.

- 5.) Run the Detailed Calibrations.
- 6.) Once Detailed Calibrations are done, re-run AUTO CT NUMBER ADJUST. Run the same system configurations that were run for Detailed Calibrations.
- 7.) Check to ensure that the CT Number Adjust scans are free of artifacts and make sure that Image Analysis now passes on the previously failed configurations.

4.13 Finalization

Save system state.